

EXPERIMENT No. 07: Caching Application Data with ElastiCache, Caching with Amazon CloudFront, Caching Strategies

7.1 Objective	7.6 Observation
7.2 Apparatus Required	7.7 Results
7.3 Pre-Requisite	7.8 Discussions
7.4 Introduction	7.9 Pre-Requisite Question
7.5 Procedure	

7.1 Objectives: Caching Application Data with ElastiCache, Caching with Amazon CloudFront, Caching Strategies

7.2 Apparatus Required:

AWS Account

7.3 Pre-Requisite:

AWS basics, caching basics and AWS CloudFront

7.4 Introduction:

AWS CloudFront - CloudFront offers several options for streaming your media to global viewers—both pre-recorded files and live events. For video on demand (VOD) streaming, you can use CloudFront to stream in common formats such as MPEG DASH, Apple HLS, Microsoft Smooth Streaming, and CMAF, to any device.

The two main components of **AWS Cloudfront** are **content delivery** and **dynamic content caching**

7.5 Procedure:

ON LEFT SIDE RESOURCES - [] REDIS OSS CACHE-- []

CONTINUE TO REDDIS OSS

Configuration info

Choose one of the following options to create a Redis cache.

Deployment option

- ☐ Serverless - new
Use to quickly create a cache that automatically scales to meet application traffic demands, with no servers to manage.
- ☒ Design your own cache
Use to create a cache by selecting node type, size, and count.

Creation method

- ☐ Easy create
Use recommended best practice configurations. You can also modify options after you create the cluster.
- ☒ Cluster cache
Set all of the configuration options for your new cluster.
- ☐ Restore from backup
Use an existing backup or .rdb file to restore a cluster.

Cluster mode

Scale your cluster dynamically with no downtime.

- ☒ Enabled
Cluster mode enables replication across multiple shards for enhanced scalability and availability.
- ☐ Disabled
The Redis OSS cluster will have a single shard (node group) with one primary node and up to 5 read replica.

Enabling cluster mode supports partitioning your data across up to 500 node groups and improves performance of Redis OSS clusters. Some commands are unavailable in this mode. [Learn more](#)

Cluster info

Use the following options to configure the cluster.

Name

The name can have up to 40 characters, and must not contain spaces.

Description - optional

Location

Choose whether to host the cluster in the AWS Cloud or on premises.

Location

☒ AWS Cloud

Use the AWS Cloud for your ElastiCache instances.

☐ On premises

Create your ElastiCache instances on an Outpost (through AWS Outposts). You need to create a subnet ID on an Outpost first.

Multi-AZ

☐ Enable

Multi-AZ provides enhanced high availability through automatic failover to a read replica, cross AZs, in case of a primary node failover.

Auto-failover

#####DON'T ENABLE MULTI A-Z

SELECT ENGINE VERSION (7.0) , PORT NUM (6379), **NODE TYPE (t3 micro)**, SHARDS 2

Cluster settings

Use the following options to configure the cluster.

Engine version

Version compatibility of the engine that will run on your nodes.

Port

The port number that nodes accept connections on.

Parameter groups

Parameter groups control the runtime properties of your nodes and clusters.



Node type

The type of node to be deployed and its associated memory size.

0.5 GiB memory Up to 5 Gigabit network performance

Number of shards

Enter the number of shards in this cluster, from 1 to 500.

Replicas per shard

Enter the number of replicas for each shard, from 0 to 5.

Connectivity

Choose the IP version(s) this cluster will support. Then select an existing subnet group or create a new one.

Network type

Choose between IPv4, dual stack and IPv6

Your resources will communicate only over the IPv4 protocol.

Subnet groups

☐ Choose existing subnet group

☒ Create a new subnet group

Name

The name can have up to 255 characters, and must not contain spaces.

Description - optional

Subnet groups

☐ Choose existing subnet group

☒ Create a new subnet group

Name

The name can have up to 255 characters, and must not contain spaces.

Description - optional

VPC ID

The identifier for the VPC environment where your cluster is to run.

[Create VPC](#)

For Multi-AZ high availability mode, choose IDs for at least two subnets from two Availability Zones in the table below.

Selected subnets (2)

[Manage](#)

Availability Zone	Subnet ID	Subnet name	CIDR block (IPv4)
eu-north-1a	subnet-01b1ab8953d783de1	-	172.31.16.0/20
eu-north-1b	subnet-0fdf29b510af5dba3	-	172.31.32.0/20

GIVE REDDIS SUBNET NAME AND MANAGE THE RESOURCE

Name

Description

VPC

Select

Availability

eu-north-1

Tags for subnet group

Manage subnets

Add or remove subnets from the table below.

Subnets (1/3)

Find subnets

Availability Zone	Subnet ID	Subnet name	CIDR block (IPv4)
☒ eu-north-1a	subnet-01b1ab8953d783de1	-	172.31.16.0/20
☐ eu-north-1b	subnet-0fdf29b510af5dba3	-	172.31.32.0/20
☐ eu-north-1c	subnet-0f92911b6a96d8908	-	172.31.0.0/20

Cancel Choose

Availability Zone placements

Use the following fields to configure placements for Availability Zones.

Slots and keyspaces

Distribution of the 16,384 cluster keyspace slots across shards.

Equal distribution

Availability Zone placements

By locating nodes in different Availability Zones, you reduce the chance that a failure in one Availability Zone, such as a power outage, will cause your entire system to fail. Choose Specify Availability Zones if you want to specify Availability Zones for cluster nodes.

Specify Availability Zones

Shards

Slots/keyspaces

Primary

Replica 1

Shard 1	Equal distribution	eu-north-1a	eu-north-1b
Shard 2	Equal distribution	eu-north-1a	eu-north-1b

Cancel Next

NEXT

Advanced settings [Info](#)

Security

Use the following section to configure network security and data security for your cluster.

Encryption at rest

☒ Enable

Enables encryption of data stored on disk.

Encryption key

The master key that will be used to protect the key used to encrypt data at rest.

☒ Default key

An AWS owned key will be used for encryption.

☐ Customer managed CMK

Select a customer managed key.

Encryption in transit [Info](#)

☒ Enable

Enables encryption of data that moves between the service and client.

Transit encryption mode

Required

☐ In Required mode, the cluster will support only encrypted TLS connections. Transit encryption mode can be modified after the cluster has been created. [Learn more](#)

Select AUTH DEFAULT ACCESS

Access control

Provides the ability to configure authenticating and authorizing access.

AUTH default user access

AUTH token

The AUTH token used for the cluster.

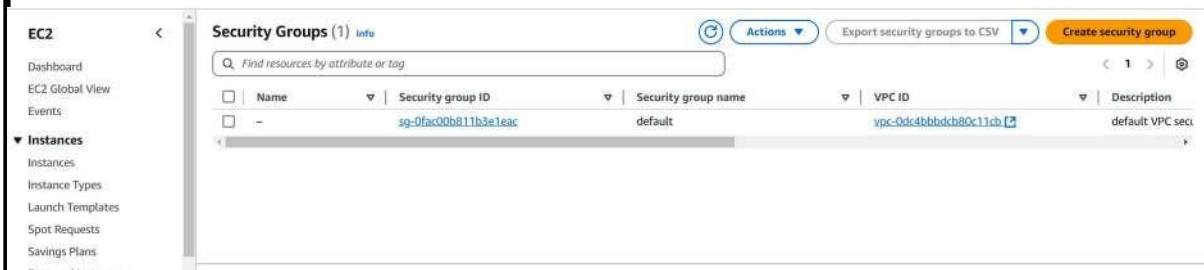
AUTHtoken123456789

☒ Show token

At least 16 characters and a maximum of 128 characters, which can be any printable ASCII character except for ' ' (blank space), "" (quotation mark), '/', and '@'.

AUTHtoken123456789

STEP2: GO TO EC2 IN NEW TAB AND CREATE A SECURITY GROUP



EC2 > Security Groups > Create security group

Create security group

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name

REDDISSEC26

Name cannot be edited after creation.

Description

Allows SSH access to developers

VPC

vpc-0dc4bbbdcb80c11cb

INBOUND RULES ☐ **ADD RULE** ☐ **CUSTOM TCP** ☐ **PORT NUM** ☐ **6379** (AS OUR ELASTIC STACK RUNNING OVER THERE) - ☐ **SOURCE** (ANYWHERE IPV4)

Inbound rules

Type

Custom TCP

Protocol

TCP

Port range

6379

Source

Anyw...

0.0.0.0/0

Description - optional

Add rule

CREATE SECURITY GROUP

Inbound rules Info

Type Info: Custom TCP | Protocol Info: TCP | Port range Info: 6379 | Source Info: Anywhere... | Description - optional Info:

Outbound rules Info

Type Info: All traffic | Protocol Info: All | Port range Info: All | Destination Info: Custom | Description - optional Info:

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

You can add up to 50 more tags.

Security group sg-08b8fd734ba812a46 (REDDISSEC26) was created successfully

sg-08b8fd734ba812a46 - REDDISSEC26

Details

Security group name sg-08b8fd734ba812a46	Security group ID sg-08b8fd734ba812a46	Description REDDISSEC26	VPC ID vpc-0d4b8bcb60c11cb17
Owner 695475937818	Inbound rules count 1 Permission entry	Outbound rules count 1 Permission entry	

Inbound rules | Outbound rules | Sharing - new | VPC associations - new | Tags

Inbound rules (1)

☐	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
☐	-	sg-05bc2623d3a61e4a0	IPv4	Custom TCP	TCP	6379	0.0.0.0/0	-

----- STEP3: COMEBACK TO ELASTIC CACHE AND SELECT THE SECURITY GROUP----

Selected security groups (0)

A security group acts like a firewall that controls network access to your clusters.

Group ID

No selected security groups

Add security groups by choosing the Manage button.

GO TO MANAGE AND REFRESH & SELECT THE SECURITY GROUP

AUTH default user access

Manage security groups

Security groups (1/2)

Find security groups

Security group ID

Security group name

Description

sg-08b8fd734ba812a46

REDDISSEC26

REDDISSEC26

sg-0fac00b811b3e1eac

default

default VPC security group

Cancel

Choose

Backup

You can use backups to restore a cluster or seed a new cluster. The backup consists of the cluster's metadata, along with all of the data in the cluster.

☐ Enable automatic backups

ElastiCache will automatically create a daily backup of a set of replicas.

Maintenance

Configure maintenance settings for the cluster.

Maintenance window

Specify the time range (UTC) for updates such as patching an operating system, updating drivers, and installing software or patches.

☒ No preference

☐ Specify maintenance window

Auto upgrade minor versions

☐ Enable

Automatically schedule cluster upgrade to the latest minor version, once it becomes available. Cluster upgrade will only be scheduled during the maintenance window.

Topic for Amazon SNS notification

Choose an SNS topic from the list, or enter the Amazon Resource Name (ARN) for an existing topic. If no topic is chosen, no notifications are sent.

Disable notifications

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8

Logs

Specify whether to provide the Redis OSS slow logs or engine logs.

Slow logs

☒ Enable

Provide the slow log for queries that exceed a specified runtime.

Log format

Choose the format of the logs.

Text

Log destination type

Choose the destination of the logs.

CloudWatch Logs

Log destination

Select an existing log group or create a new one.

☐ Choose existing log group

☒ Create a new log group

Log group name

REDIS26LOG

Engine logs

☐ Enable

Provide the engine log for queries that exceed a specified runtime.

Tags

You can use tags to search and filter your clusters, or track your AWS costs.

No tags associated with the cluster.

[Add new tag](#)

You can add 50 more tags.

[Cancel](#)

[Previous](#)

[Next](#)

NEXT

WITH DEFAULT SETTINGS IN NEXT PAGE CLICK ON
CREATE

Logs

Specify whether to provide the Redis OSS slow logs or engine logs.

Slow logs

Enabled

Slow logs format

text

Slow log destination type

cloudwatch-logs

Slow log destination

REDIS26LOG [✕](#)

Engine logs

Disabled

Tags

You can use tags to search and filter your clusters, or track your AWS costs.

Key

Value

No tags found.

Tags that you create will appear here.

[Cancel](#)

[Previous](#)

[Create](#)

ElastiCache > Redis OSS caches

The cluster was created successfully.

Redis OSS caches (1) [Info](#)
[View details](#)
[Actions](#)
[Create Redis OSS cache](#)

Cache name	Status	Description	Engine version	Configuration	Created date
myreddis26	Creating	-	7.0.7	cache.t3.micro	February 5, 2025, 03:5...

EXPERIMENT No. 08: Implementing CloudFront for Caching and Application Security

8.1 Objective	8.6 Observation
8.2 Apparatus Required	8.7 Results
8.3 Pre-Requisite	8.8 Discussions
8.4 Introduction	8.9 Pre-Requisite Question
8.5 Procedure	

8.1 Objectives: Implementing CloudFront for Caching and Application Security

8.2 Apparatus Required:

AWS Account

8.3 Pre-Requisite:

AWS Basics, CloudFront knowledge

8.4 Introduction:

Amazon CloudFront is a content delivery network (CDN) operated by Amazon Web Services. The content delivery network was created to provide a globally distributed network of proxy servers to cache content, such as web videos or other bulky media, more locally to consumers, to improve access speed for downloading the content.

8.5 Procedure:

GO TO S3 & CREATE A BUCKET

GIVE BUCKET NAME AND KEEP ALL DEFAULTS

Amazon S3

Buckets

Create bucket

Search

[Alt+5]

Europe (Stockholm)

Idponco

Create bucket Info

Buckets are containers for data stored in S3.

General configuration

AWS Region
Europe (Stockholm) eu-north-1

Bucket type Info

☒ General purpose
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ Directory
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket name Info

Bucket name must be unique within the global namespace and follow the bucket naming rules. [See rules for bucket naming](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

Format: s3://bucket/prefix

Object Ownership Info

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

☒ Object ownership enforced by Amazon S3

☐ Manage ACLs

CloudShell

Feedback

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Object Ownership
Bucket owner enforced

Block Public Access settings for this bucket

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

☒ Block all public access
Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

☒ Block public access to buckets and objects granted through new access control lists (ACLs)
S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.

☒ Block public access to buckets and objects granted through any access control lists (ACLs)
S3 will block all public access permissions applied to any bucket or object, regardless of whether the permissions were granted through new or existing ACLs.

☒ Block public access to buckets and objects granted through new public bucket or access point policies
S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.

☒ Block public and cross-account access to buckets and objects through any public bucket or access point policies
S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

Bucket Versioning

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning

☐ Disable

☒ Enable

Tags - optional (0)

You can use bucket tags to track storage costs and organize buckets. [Learn more](#)

No tags associated with this bucket.

Default encryption Info

Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type Info

☒ Server-side encryption with Amazon S3 managed keys (SSE-S3)

☐ Server-side encryption with AWS Key Management Service keys (SSE-KMS)

☐ Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)
Secure your objects with two separate layers of encryption. For details on pricing, see [DSSE-KMS pricing on the Storage tab of the Amazon S3 pricing page](#).

Bucket Key
Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS. [Learn more](#)

☐ Disable

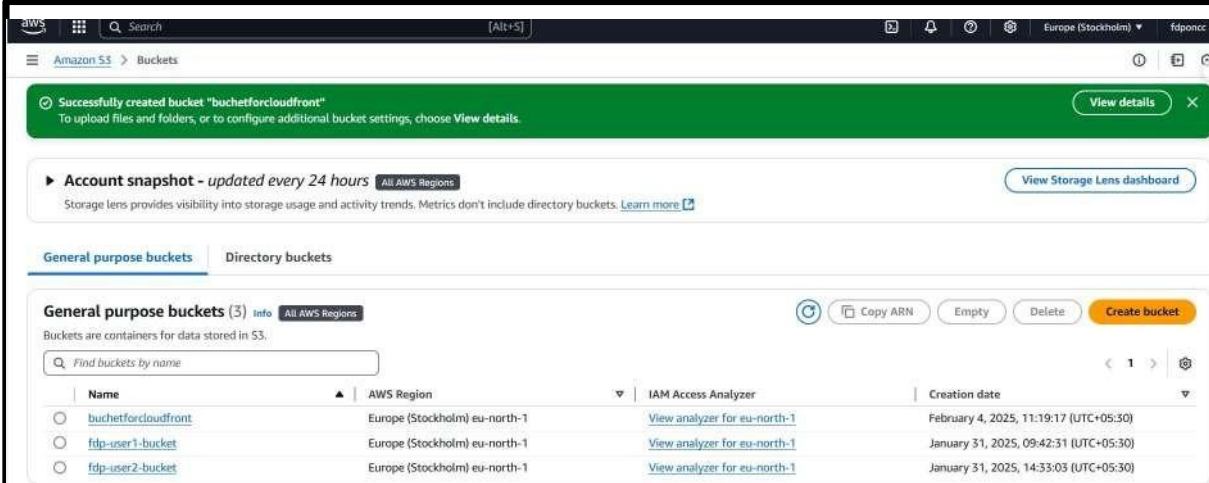
☒ Enable

Advanced settings

After creating the bucket, you can upload files and folders to the bucket, and configure additional bucket settings.

Cancel

Create bucket



Successfully created bucket "bucketforcloudfront"
To upload files and folders, or to configure additional bucket settings, choose [View details](#).

Account snapshot - updated every 24 hours [All AWS Regions](#) [View Storage Lens dashboard](#)
Storage lens provides visibility into storage usage and activity trends. Metrics don't include directory buckets. [Learn more](#)

General purpose buckets | Directory buckets

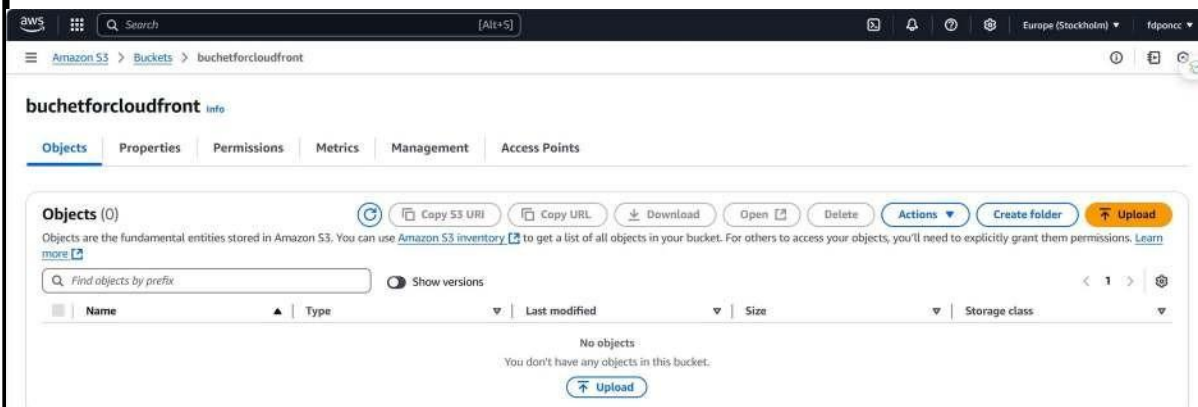
General purpose buckets (3) [info](#) [All AWS Regions](#)

Buckets are containers for data stored in S3.

Find buckets by name

Name	AWS Region	IAM Access Analyzer	Creation date
bucketforcloudfront	Europe (Stockholm) eu-north-1	View analyzer for eu-north-1	February 4, 2025, 11:19:17 (UTC+05:30)
fdp-user1-bucket	Europe (Stockholm) eu-north-1	View analyzer for eu-north-1	January 31, 2025, 09:42:31 (UTC+05:30)
fdp-user2-bucket	Europe (Stockholm) eu-north-1	View analyzer for eu-north-1	January 31, 2025, 14:33:03 (UTC+05:30)

GO TO BUCKET CREATED



bucketforcloudfront [info](#)

Objects | Properties | Permissions | Metrics | Management | Access Points

Objects (0) [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open](#) [Delete](#) [Actions](#) [Create folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

Show versions

Name	Type	Last modified	Size	Storage class
No objects You don't have any objects in this bucket.				

[Upload](#)

GO TO PROPERTIES AND HOST STATIC WEBSITE

Edit static website hosting [info](#)

Static website hosting

Use this bucket to host a website or redirect requests. [Learn more](#)

Static website hosting

- ☐ Disable
☒ Enable

Hosting type

- ☒ Host a static website
Use the bucket endpoint as the web address. [Learn more](#)
- ☐ Redirect requests for an object
Redirect requests to another bucket or domain. [Learn more](#)

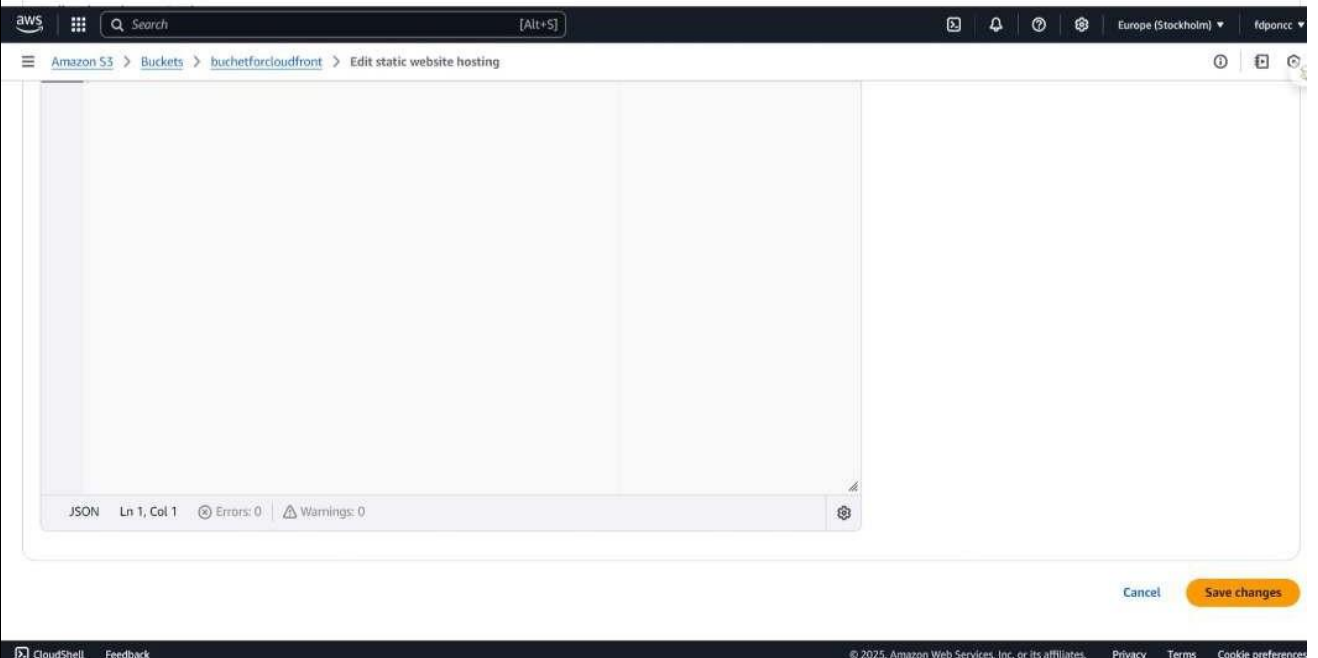
For your customers to access content at the website endpoint, you must make all your content publicly readable. To do so, you can edit the S3 Block Public Access settings for the bucket. For more information, see [Using Amazon S3 Block Public Access](#)

Index document

Specify the home or default page of the website.

Error document - optional

This is returned when an error occurs.



The screenshot shows the AWS Management Console interface for editing static website hosting. The top navigation bar includes the AWS logo, a search bar, and the user's profile. The breadcrumb trail indicates the path: Amazon S3 > Buckets > bucketforcloudfront > Edit static website hosting. The main content area is a large, empty text box for editing the index document and error document. At the bottom right of the console, there are 'Cancel' and 'Save changes' buttons. The footer of the console shows 'CloudShell', 'Feedback', and copyright information for Amazon Web Services, Inc. or its affiliates.

SAVE CHANGES

GO TO OBJECT AND UPLOAD A FILE (HTML)



Search

[Alt+S]

Europe (Stockholm)

Idponcz

Amazon S3

Buckets

bucketforcloudfront

Upload

Upload

info

Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. [Learn more](#)

Drag and drop files and folders you want to upload here, or choose [Add files](#) or [Add folder](#).

Files and folders (1 total, 296.0 B)

Remove

Add files

Add folder

All files and folders in this table will be uploaded.

Find by name

☐

Name

☐

second.html

Folder

-

Type

text/html

Size

296.0 B

Destination

info

Destination

s3://bucketforcloudfront

Destination details

Bucket settings that impact new objects stored in the specified destination.

Permissions

Grant public access and access to other AWS accounts.

CloudShell

Feedback

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Properties

Specify storage class, encryption settings, tags, and more.

Storage class

info

Amazon S3 offers a range of storage classes designed for different use cases. [Learn more](#) or see [Amazon S3 pricing](#)

	Storage class	Designed for	Bucket type	Availability Zones	Min storage duration	Min billable object size	Monitoring and auto-tiering fees	Retrieval fees
☒	Standard	Frequently accessed data (more than once a month) with milliseconds access	General purpose	≥ 3	-	-	-	-
☐	Intelligent-Tiering	Data with changing or unknown access patterns	General purpose	≥ 3	-	-	Per-object fees apply for objects ≥ 128 KB	-
☐	Standard-IA	Infrequently accessed data (once a month) with milliseconds access	General purpose	≥ 3	30 days	128 KB	-	Per-GB fees apply
☐	One Zone-IA	Recreatable, infrequently accessed data (once a month) with milliseconds access	General purpose or directory	1	30 days	128 KB	-	Per-GB fees apply
☐	Glacier Instant Retrieval	Long-lived archive data accessed once a quarter with instant retrieval in milliseconds	General purpose	≥ 3	90 days	128 KB	-	Per-GB fees apply
☐	Glacier Flexible Retrieval (formerly Glacier)	Long-lived archive data accessed once a year with retrieval of minutes to hours	General purpose	≥ 3	90 days	-	-	Per-GB fees apply
☐	Glacier Deep Archive	Long-lived archive data accessed less than once a year with retrieval of hours	General purpose	≥ 3	180 days	-	-	Per-GB fees apply
☐	Reduced redundancy	Noncritical, frequently accessed data with milliseconds access (not recommended as S3 Standard is more cost effective)	General purpose	≥ 3	-	-	-	Per-GB fees apply

Server-side encryption

info

Server-side encryption protects data at rest.

Server-side encryption

☒

Don't specify an encryption key

The bucket settings for default encryption are used to encrypt objects when storing them in Amazon S3.

☐

Specify an encryption key

The specified encryption key is used to encrypt objects before storing them in Amazon S3.

⚠ If your bucket policy requires objects to be encrypted with a specific encryption key, you must specify the same encryption key when you upload objects. Otherwise, uploads will fail.

Checksums

Checksums are used for data integrity verification of new objects. [Learn more](#)

Checksum function

Checksum functions are used to calculate the checksum value. For objects smaller than 16 MB, only the full object checksum type is supported, for all checksum algorithms.

CRC64NVME (recommended)

Precalculated value - optional

When you provide a precalculated value for a single object, S3 compares it to the value it calculates using the selected checksum function. If the values don't match, the upload will fail. [Learn more](#)

Enter value

The precalculated value must be a Base64 encoded string. It must not exceed 128 characters, and can contain only letters (a-z, A-Z), numbers (0-9), forward slash (/), plus (+), or equals (=).

Tags - optional

You can use object tags to analyze, manage, and specify permissions for objects. [Learn more](#)

No tags associated with this resource.

Add tag

Metadata - optional

Metadata is optional information provided as a name-value (key-value) pair. [Learn more](#)

No metadata associated with this resource.

Add metadata

Dept. of CSE, RIT, HASSAN

15

Upload: status

Upload succeeded
For more information, see the Files and folders table.

Summary

Destination	Succeeded	Failed
s3://bucketforcloudfront	1 file, 296.0 B (100.00%)	0 files, 0 B (0%)

Files and folders (1 total, 296.0 B)

Name	Folder	Type	Size	Status	Error
second.html		text/html	296.0 B	Succeeded	

ACCESS URL FROM UR BUCKET -> PROPERTIES-> STATIC WEB HOSTING > SITE ADDRESS

Static website hosting

Use this bucket to host a website or redirect requests. [Learn more](#)

We recommend using AWS Amplify Hosting for static website hosting
Deploy a fast, secure, and reliable website quickly with AWS Amplify Hosting. [Learn more about Amplify Hosting](#) or [View your existing Amplify apps](#)

S3 static website hosting
Enabled

Hosting type
Bucket hosting

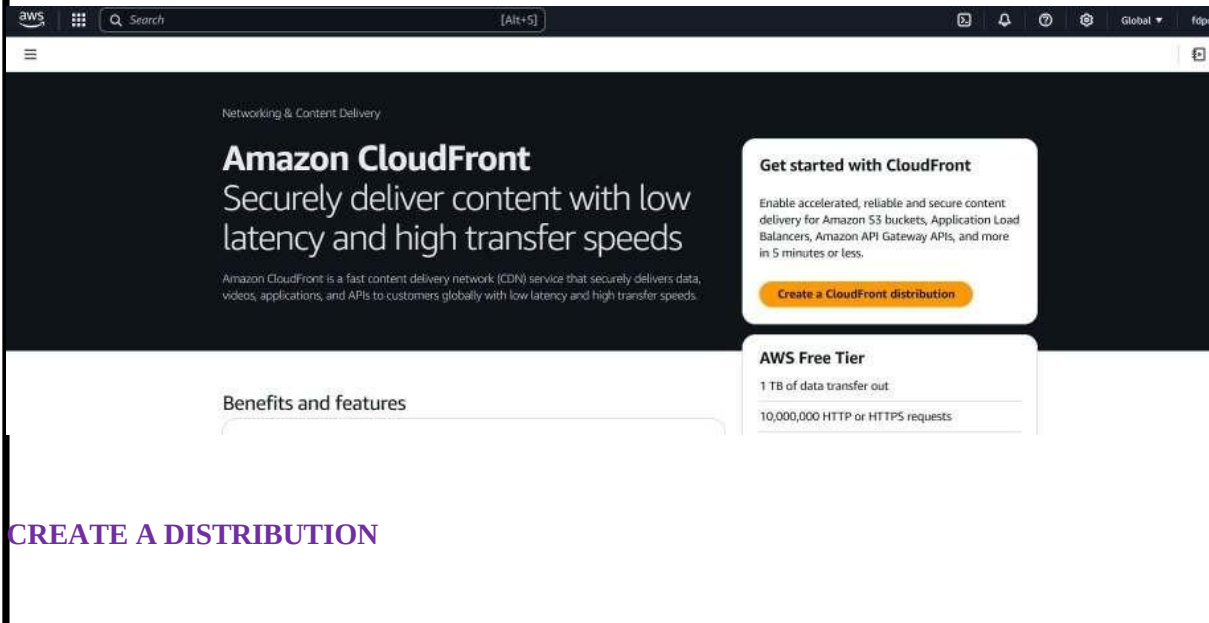
Bucket website endpoint
When you configure your bucket as a static website, the website is available at the AWS Region-specific website endpoint of the bucket. [Learn more](#)
<http://bucketforcloudfront.s3-website.eu-north-1.amazonaws.com>

IF YOU TRY TO ACCESS THE URL

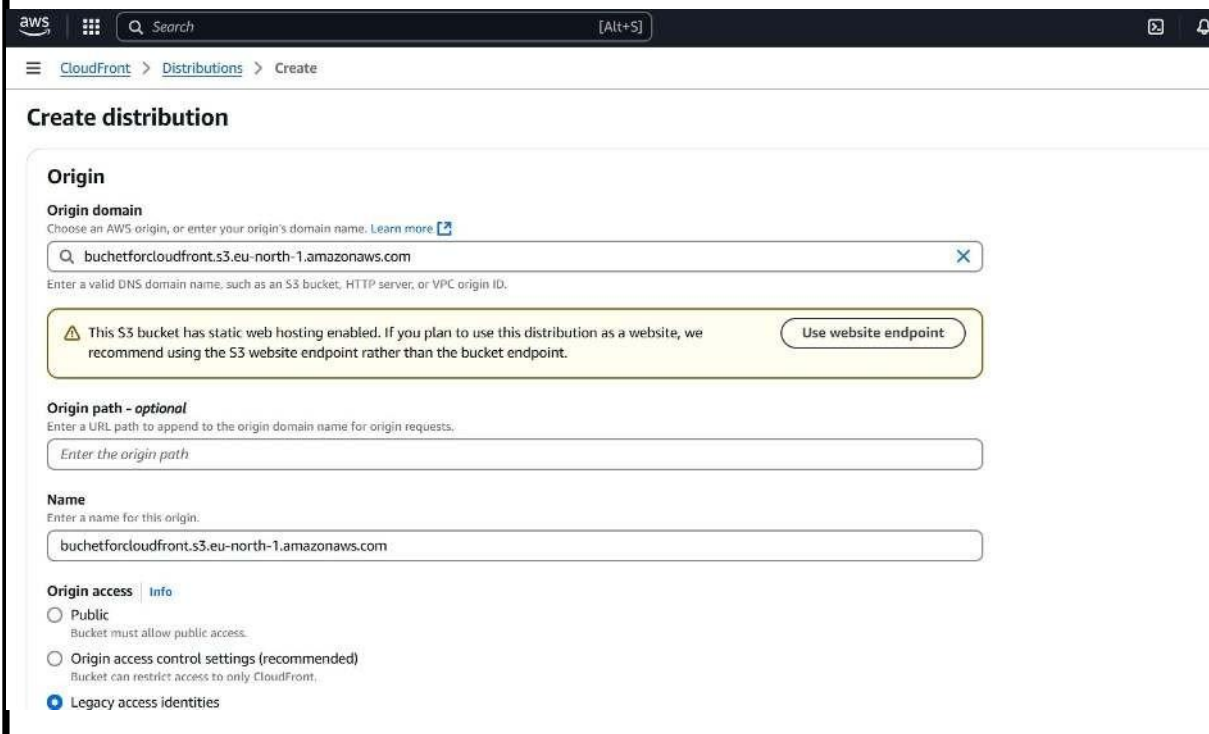
403 Forbidden

- Code: AccessDenied
- Message: Access Denied
- RequestId: YAF7TX6H40F421D2
- HostId: Irkdb7M7HGG4X26oginZ7YKa1+YaRcxnsC5sqVmyfsrYiuWoKLPmCM9jb9JRyW3Fb019ORyVWQ/P592gPrCw==

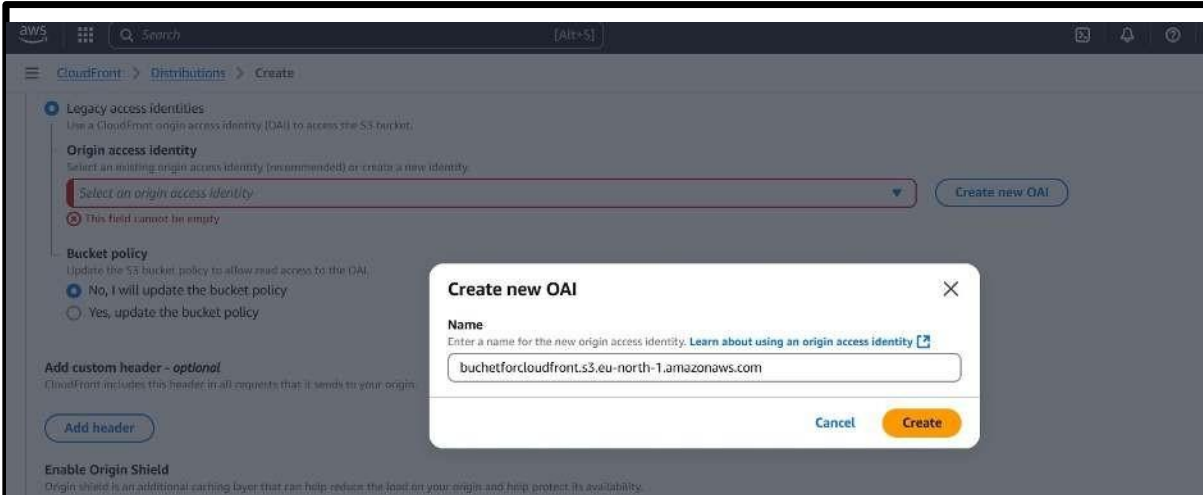
STEP 2: CLOUDFRONT



CREATE A DISTRIBUTION



CREATE A NEW OAI



aws [Search] [Alt+S]

CloudFront > Distributions > Create

Legacy access identities
Use a CloudFront origin access identity (OAI) to access the S3 bucket.

Origin access identity
Select an existing origin access identity (recommended) or create a new identity.

Select an origin access identity

Create new OAI

Bucket policy
Update the S3 bucket policy to allow read access to the OAI.

☒ No, I will update the bucket policy.
☐ Yes, update the bucket policy.

Add custom header - optional
CloudFront includes this header in all requests that it sends to your origin.

Add header

Enable Origin Shield
Origin shield is an additional caching layer that can help reduce the load on your origin and help protect its availability.

Create new OAI

Name
Enter a name for the new origin access identity. [Learn about using an origin access identity](#)

buketforcloudfront.s3.eu-north-1.amazonaws.com

Cancel Create

buketforcloudfront.s3.eu-north-1.amazonaws.com

Origin access Info

☐ Public
Bucket must allow public access.

☐ Origin access control settings (recommended)
Bucket can restrict access to only CloudFront.

☒ **Legacy access identities**
Use a CloudFront origin access identity (OAI) to access the S3 bucket.

Origin access identity
Select an existing origin access identity (recommended) or create a new identity.

buketforcloudfront.s3.eu-north-1.amazonaws.com

Create new OAI

Bucket policy
Update the S3 bucket policy to allow read access to the OAI.

☐ No, I will update the bucket policy
☒ Yes, update the bucket policy

Add custom header - optional
CloudFront includes this header in all requests that it sends to your origin.

Add header

KEEP ALL SETTINGS DEFAULT

Create distribution

Origin

Origin domain

Choose an S3 origin, or enter your origin's domain name. [Learn more](#)

Q

bucketforcloudfront.s3.eu-north-1.amazonaws.com

X

Enter a valid DNS domain name, such as an S3 bucket, HTTP server, or VPC origin ID.

This S3 bucket has static web hosting enabled. If you plan to use this distribution as a website, we recommend using the S3 website endpoint rather than the bucket endpoint.

Use website endpoint

Origin path - optional

Enter a URL path to append to the origin domain name for origin requests.

Enter the origin path

Name

Enter a name for this origin.

bucketforcloudfront.s3.eu-north-1.amazonaws.com

Origin access

Info

☐ Public

Bucket must allow public access.

☐ Origin access control settings (recommended)

Bucket can restrict access to only CloudFront.

☒ Legacy access identities

Use a CloudFront origin access identity (OAI) to access the S3 bucket.

Origin access identity

Use an existing origin access identity (recommended) or create a new identity.

bucketforcloudfront.s3.eu-north-1.amazonaws.com

Create new OAI

Bucket policy

Update the S3 bucket policy to allow read access to the OAI.

☐ No, I will update the bucket policy

☒ Yes, update the bucket policy

Add custom header - optional

CloudFront includes this header in all requests that it sends to your origin.

Add header

Enable Origin Shield

Origin shield is an additional caching layer that can help reduce the load on your origin and help protect its availability.

☒ No

☐ Yes

Additional settings

Default cache behavior

Path pattern

Info

Default (*)

Compress objects automatically

Info

☐ No

☒ Yes

Viewer

Viewer protocol policy

☒ HTTP and HTTPS

☐ Redirect HTTP to HTTPS

☐ HTTPS only

Allowed HTTP methods

☒ GET, HEAD

☐ GET, HEAD, OPTIONS

☐ GET, HEAD, OPTIONS, PUT, POST, PATCH, DELETE

Restrict viewer access

If you restrict viewer access, viewers must use CloudFront signed URLs or signed cookies to access your content.

☐ No

☒ Yes

Cache key and origin requests

We recommend using a cache policy and origin request policy to control the cache key and origin requests.

Cache policy and origin request policy (recommended)

Legacy cache settings

Cache policy

Choose an existing cache policy or create a new one.

CachingOptimized

Policy with caching enabled. Supports Gzip and Brotli compression.

Create cache policy

View policy

Origin request policy - optional

Choose an existing origin request policy or create a new one.

Select origin policy

Create origin request policy

Response headers policy - optional

Choose an existing response headers policy or create a new one.

Select response headers

Create response headers policy

Additional settings

Function associations - optional

Choose an edge function to associate with this cache behavior, and the CloudFront event that invokes the function.

Viewer request

Viewer response

Origin request

Origin response

Function type

No association

No association

No association

No association

Function ARN / Name

Include body

Web Application Firewall (WAF)

Info

☐ Enable security protections

Keep your application secure from the most common web threats and security vulnerabilities using AWS WAF. Blocked requests are stopped before they reach your web servers.

☐ Do not enable security protections

Select this option if your application does not need security protections from AWS WAF.

Included security protections

Protect against the most common vulnerabilities found in web applications.

Protect against malicious actors discovering application vulnerabilities.

Block IP addresses from potential threats based on Amazon internal threat intelligence

Price estimate

This AWS WAF configuration is estimated to cost \$14 for 10 million requests/month

Settings

Anycast static IP list - optional

Info

Receive traffic from a small set of IP addresses

There are no Anycast static IP lists available

Create an Anycast static IP list

There are no Anycast static IP lists available

Price class

Info

Choose the price class associated with the maximum price that you want to pay.

☒ Use all edge locations (best performance)

☐ Use only North America and Europe

☐ Use North America, Europe, Asia, Middle East, and Africa

Alternate domain name (CNAME) - optional

Add the custom domain names that you use in URLs for the files served by this distribution.

Add item

To add a list of alternative domain names, use the bulk editor.

Custom SSL certificate - optional

Associate a certificate from AWS Certificate Manager. The certificate must be in the US East (N. Virginia) Region (us-east-1).

Choose certificate

CloudFront > Distributions > Create

Info

Most support for Amazon EC2 instances, on-premises servers, and other services by default.

☒ HTTP/2

☐ HTTP/3

Default root object - optional

The object (file name) to return when a viewer requests the root URL (/) instead of a specific object.

IP v6

☐ Off

☒ On

Description - optional

Standard logging

Info

Additional charges may apply. See info for more details.

Log delivery

Info

Most support for Amazon EC2 instances, on-premises servers, and other services by default.

☒ Off

☐ On

Cancel

Create distribution

Default root object - optional

The object (file name) to return when a viewer requests the root URL (/) instead of a specific object.

IPv6

☐ Off
☒ On

Description - optional

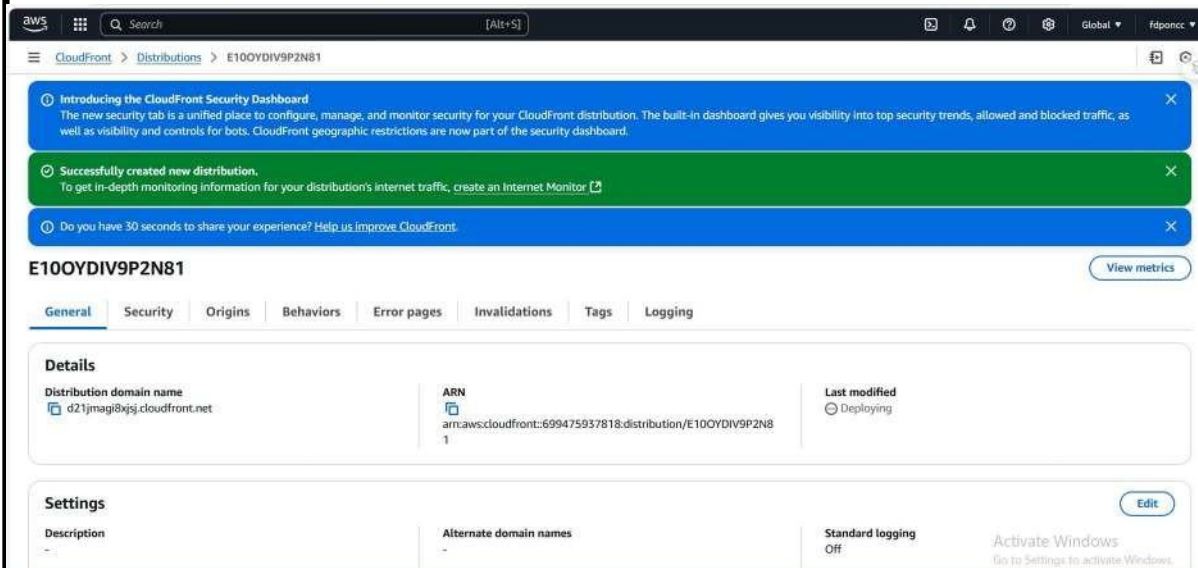
Standard logging Info

Additional charges may apply. See [Info](#) for more details.

Log delivery

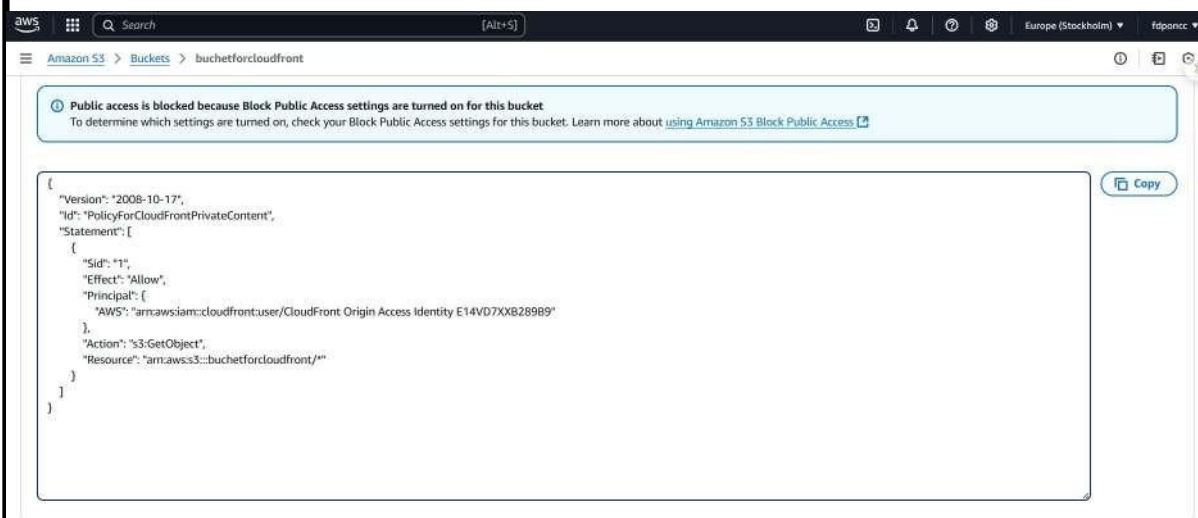
Get logs of viewer requests to CloudWatch, Amazon S3 or Firehose

☒ Off
☐ On

[Cancel](#)
[Create distribution](#)


The screenshot shows the AWS CloudFront console. At the top, there's a navigation bar with the AWS logo, a search bar, and user information. Below the navigation bar, the breadcrumb trail is "CloudFront > Distributions > E100YDIV9P2N81". There are three notification banners: a blue one about the CloudFront Security Dashboard, a green one about a successfully created new distribution, and a blue one asking for feedback. Below the notifications, the distribution name "E100YDIV9P2N81" is displayed with a "View metrics" button. The "General" tab is selected, showing details like the distribution domain name "d21jmagi8xjsj.cloudfront.net", the ARN "arn:aws:cloudfront::699475937818:distribution/E100YDIV9P2N81", and the last modified status "Deploying". The "Settings" section shows "Standard logging" as "Off" and "Alternate domain names" as empty. There's also a link to "Activate Windows".

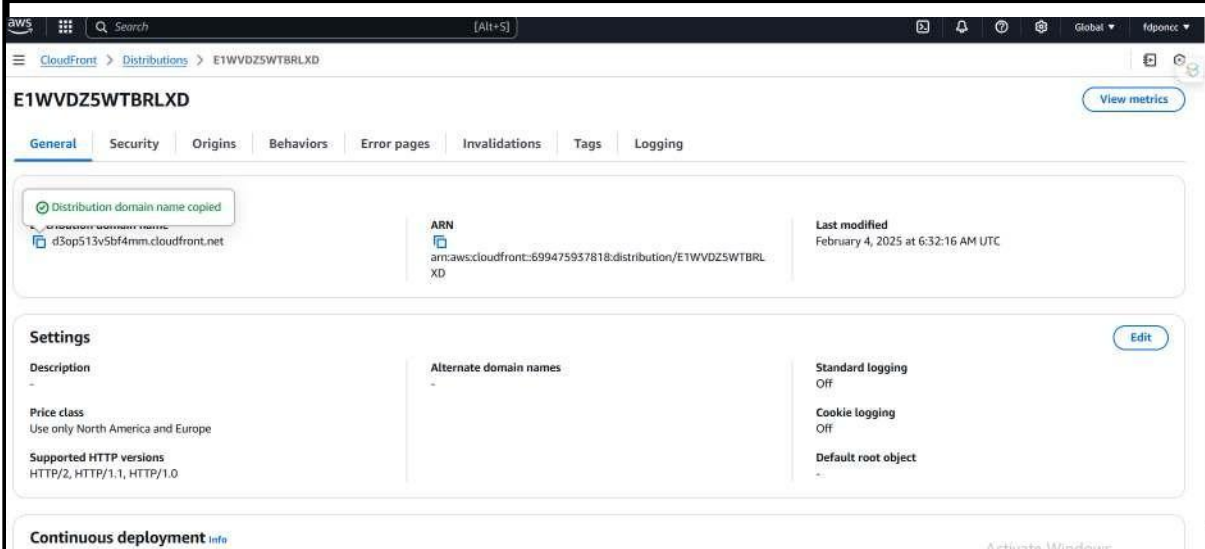
POLICY BEEN CREATED IN S3 BUCKET



The screenshot shows the AWS S3 console. The breadcrumb trail is "Amazon S3 > Buckets > bucketforcloudfront". A notification banner states: "Public access is blocked because Block Public Access settings are turned on for this bucket. To determine which settings are turned on, check your Block Public Access settings for this bucket. Learn more about [using Amazon S3 Block Public Access](#)". Below the notification, the bucket policy is displayed in a code editor. The policy is a JSON document that allows the CloudFront Origin Access Identity to get objects from the bucket. A "Copy" button is visible next to the policy text.

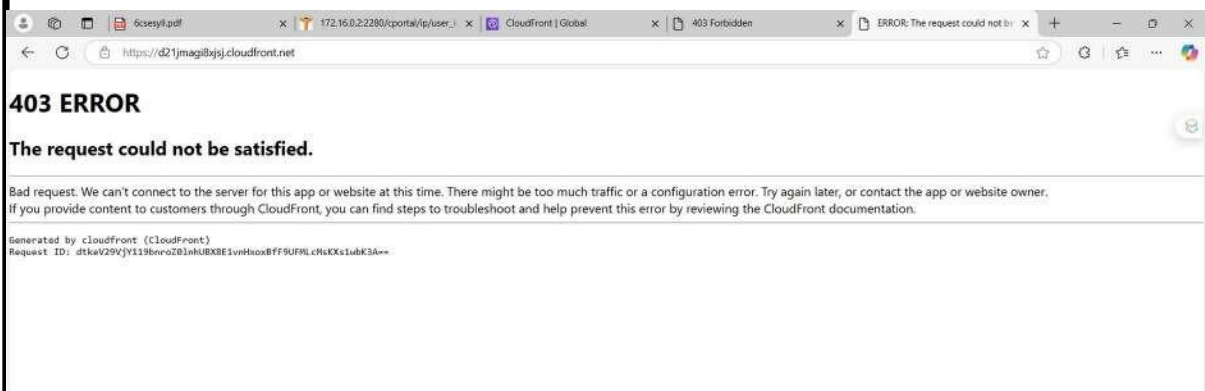
```
{
  "Version": "2008-10-17",
  "Id": "PolicyForCloudFrontPrivateContent",
  "Statement": [
    {
      "Sid": "1",
      "Effect": "Allow",
      "Principal": {
        "AWS": "arn:aws:iam::cloudfrontuser:CloudFront Origin Access Identity E14VD7XXB289B9"
      },
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::bucketforcloudfront/*"
    }
  ]
}
```

GO TO THE LINK (DISTRIBUTION DOMAIN NAME)



The screenshot shows the AWS CloudFront console for distribution E1WVDZ5WTBRLXD. The 'General' tab is selected, displaying the distribution domain name d3op513v5bf4mm.cloudfront.net, its ARN, and the last modified date. Below this, the 'Settings' section shows the description, price class, supported HTTP versions, alternate domain names, standard logging, cookie logging, and default root object. A 'Continuous deployment' section is also visible at the bottom.

S3 RESULT



The screenshot shows a web browser displaying a 403 Forbidden error. The address bar shows the URL https://d21jmagi@xjsj.cloudfront.net. The error message states: '403 ERROR The request could not be satisfied. Bad request. We can't connect to the server for this app or website at this time. There might be too much traffic or a configuration error. Try again later, or contact the app or website owner. If you provide content to customers through CloudFront, you can find steps to troubleshoot and help prevent this error by reviewing the CloudFront documentation.' The request ID is also displayed at the bottom.

CLOUD FRONT RESULT

EXPERIMENT No.09: Orchestrating Serverless Functions with AWS Step Functions

9.1 Objective	9.6 Observation
9.2 Apparatus Required	9.7 Results
9.3 Pre-Requisite	9.8 Discussions
9.4 Introduction	9.9 Pre-Requisite Question
9.5 Procedure	

9.1 Objectives: Orchestrating Serverless Functions with AWS Step Functions

9.2 Apparatus Required:

AWS Account

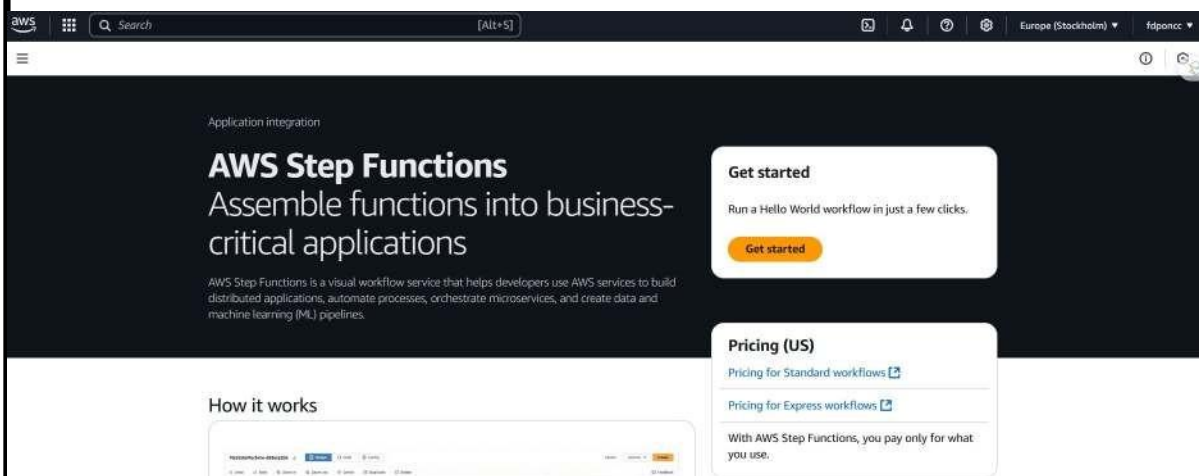
9.3 Pre-Requisite:

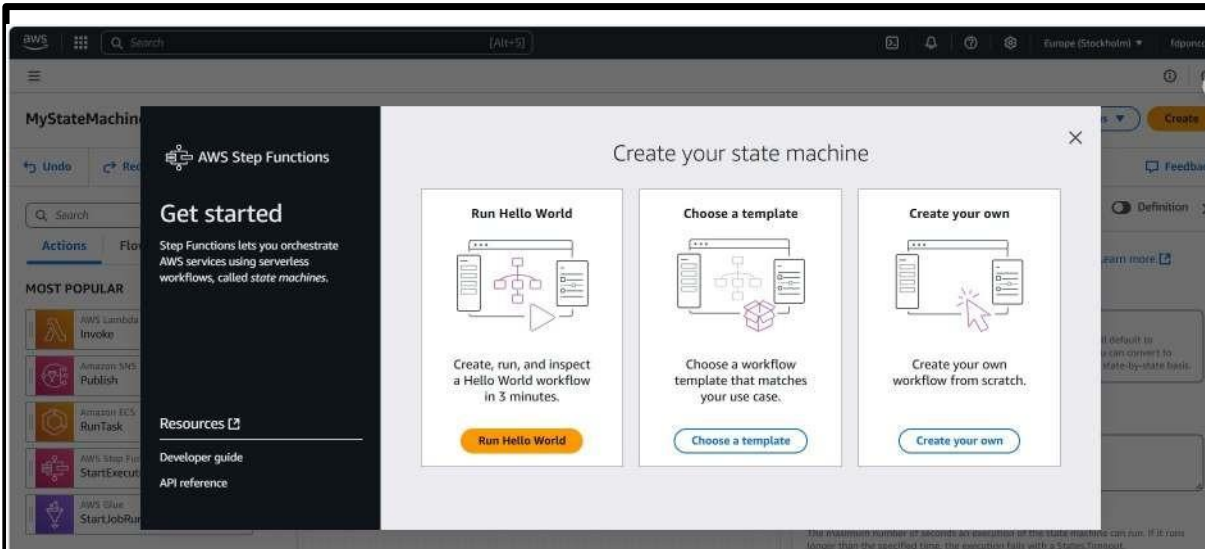
AWS basics, AWS step functions

9.4 Introduction:

Serverless functions let developers code in any language or framework with which they're comfortable. Simpler backend code. Serverless removes a lot of coding complexity for developers, allowing them to create simple, self-contained functions that independently perform one purpose.

GO TO STEP FUNCTION CONSOLE





CREATE YOUR OWN

#####

Lab 1. Step Function to Select/Reject a student based on Maths/Physics cut-off mark

#####

Task 1: Create SNS topic

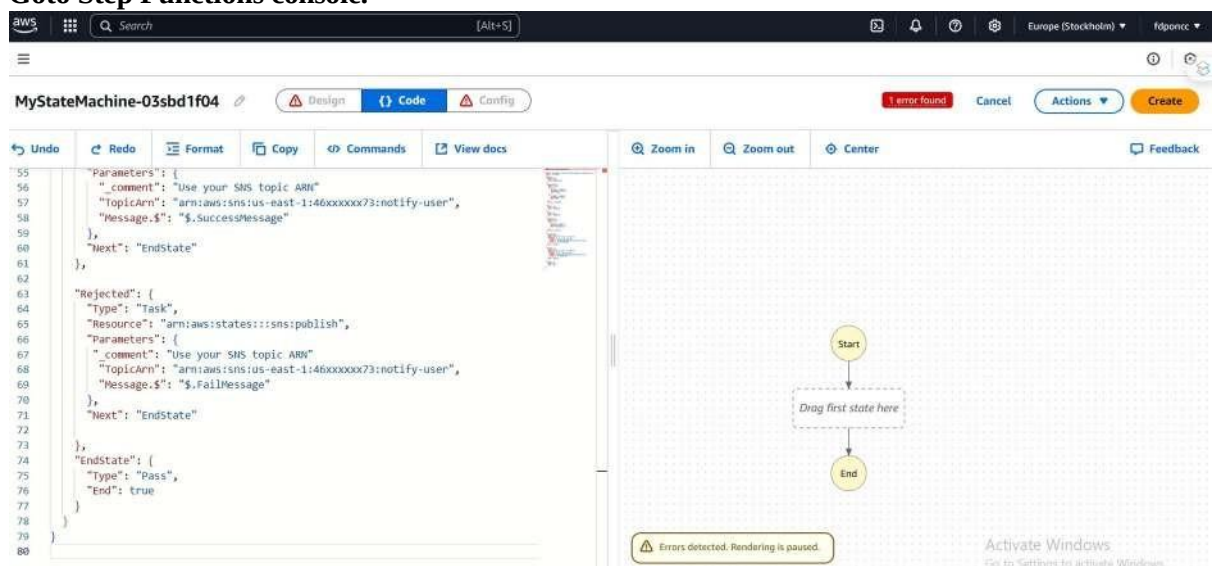
=====

1. Create an SNS topic. Subscribe your mail id to the topic.

Task 2: Definition of Step Function

=====

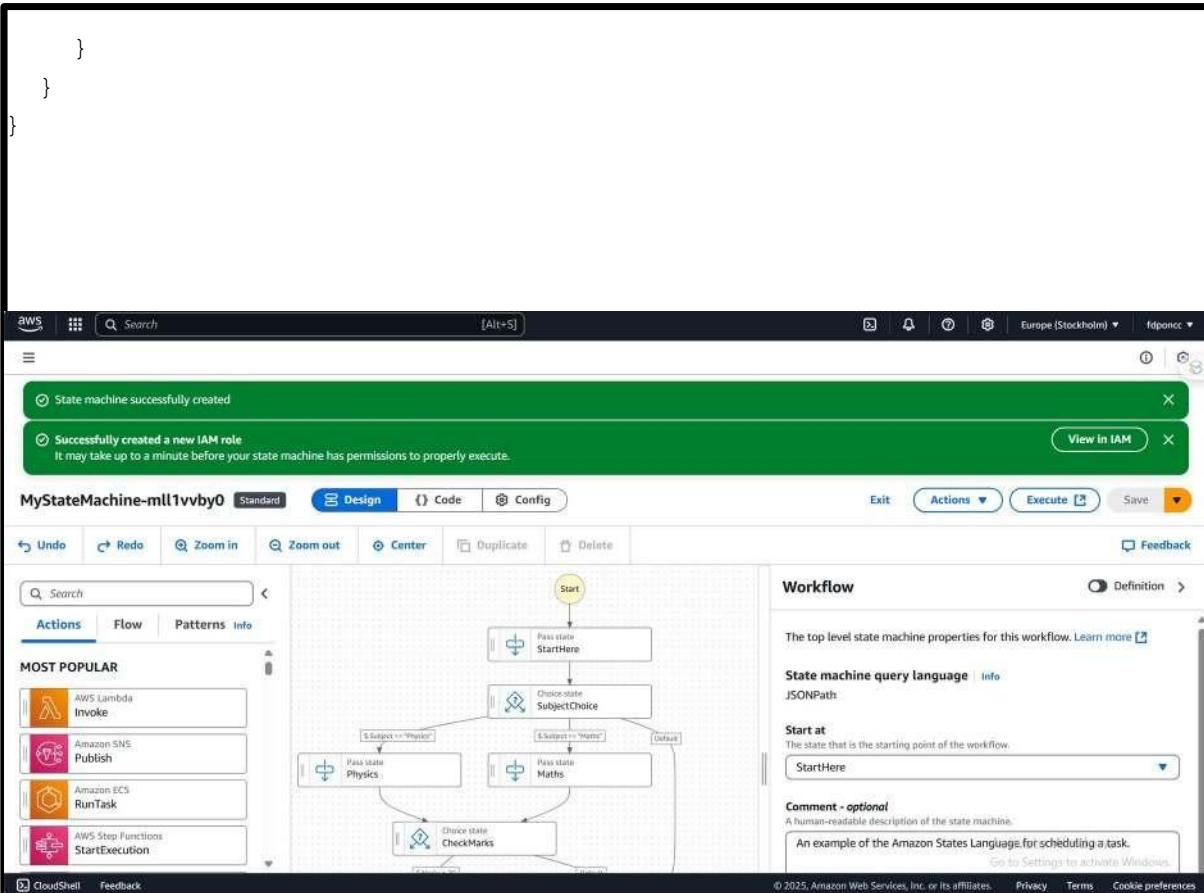
1. Goto Step Functions console.



Create a State Machine. Use the below code.

Name: `course_selection_state_machine`

```
{
  "Comment": "An example of the Amazon States Language for scheduling a
task.",
  "StartAt": "StartHere",
  "States": {
    "StartHere": {
      "Type": "Pass",
      "Next": "SubjectChoice"
    },
    "SubjectChoice": {
      "Type": "Choice",
      "Choices": [
        {
          "Variable": "$.Subject",
          "StringEquals": "Physics",
          "Next": "Physics"
        },
        {
          "Variable": "$.Subject",
          "StringEquals": "Maths",
          "Next": "Maths"
        }
      ],
      "Default": "EndState"
    },
    "Physics": {
      "Type": "Pass",
      "Next": "CheckMarks"
    },
    "Maths": {
      "Type": "Pass",
      "Next": "CheckMarks"
    },
    "CheckMarks": {
      "Type": "Choice",
      "Choices": [
        {
          "Variable": "$.Marks",
          "NumericGreaterThan": 70,
          "Next": "EndState"
        }
      ],
      "Default": "EndState"
    },
    "EndState": {
      "Type": "Pass",
      "End": true
    }
  }
}
```



2. Execute the state machine. Provide the below timer value as the input.

Execution input:

```
{
```

"Subject": "Maths",

"Marks": 76

```
}
```

```
{
```

"Subject": "Physics",

"Marks": 91

```
}
```

3. Observe the graph view. Observe each step turning green.

4. Cleanup - delete the step function

```
#####
```

```
End
```

```
#####
```

EXPERIMENT No. 10: Automating Application Deployment Using CI/CD Pipeline

10.1 Objective	10.6 Observation
10.2 Apparatus Required	10.7 Results
10.3 Pre-Requisite	10.8 Discussions
10.4 Introduction	10.9 Pre-Requisite Question
10.5 Procedure	

9.1 Objectives: Automating Application Deployment Using CI/CD Pipeline

9.2 Apparatus Required:

AWS Account

9.3 Pre-Requisite:

AWS knowledge, basic CI/CD knowledge, IAM Roles, AWS Codedeploy

9.4 Introduction:

Serverless functions let developers code in any language or framework with which they're comfortable. Simpler backend code. Serverless removes a lot of coding complexity for developers, allowing them to create simple, self-contained functions that independently perform one purpose.

A continuous integration and continuous deployment (CI/CD) pipeline is a series of steps that must be performed in order to deliver a new version of software. CI/CD pipelines are a practice focused on improving software delivery throughout the software development life cycle via automation

AWS IAM - AWS Identity and Access Management (IAM) is a web service for securely controlling access to AWS services. With IAM, you can centrally manage users, security credentials such as access keys, and permissions that control which AWS resources users and applications can access.

AWS CodeDeploy is a service that automates code deployments to any instance, including Amazon EC2 instances and instances running on-premises. AWS CodeDeploy makes it easier for you to rapidly release new features, helps you avoid downtime during deployment, and handles the complexity of updating your applications.

Step 1: Create IAM Role for EC2 and AWS CodeDeploy

- Navigate to IAM service.
- Then go to roles and create a new role.
- Select trusted entity type as AWS Service and use case as EC2

Select trusted entity Info

Trusted entity type

☒ **AWS service**
Allow AWS services like EC2, Lambda, or others to perform actions in this account.

☐ **AWS account**
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.

☐ **SAML 2.0 federation**
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.

☐ **Custom trust policy**
Create a custom trust policy to enable others to perform actions in this account.

Use case

Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case

EC2

Choose a use case for the specified service.

Use case

☒ **EC2**
Allows EC2 instances to call AWS services on your behalf.

☐ **EC2 Role for AWS Systems Manager**

Step 2: Add permissions To IAM Role

- **Select AmazonS3ReadOnlyAccess permission. It will allow our EC2 instance to access stored artifacts from the Amazon S3 bucket.**

Add permissions Info

Permissions policies (1/884) Info

Choose one or more policies to attach to your new role.

× Filter All

☒	Policy name ↗	Type
☒	  AmazonS3ReadOnlyAccess	AWS manage

► Set permissions boundary - *optional*

Step 3: Creating The Role For AWS CodeDeploy

- Provide the Name, review and Click on Create for creating the Role.
- Select an appropriate role name and click on create role.

Name, review, and create

Role details

Role name
Enter a meaningful name to identify this role.

Maximum 64 characters. Use alphanumeric and '+=, @-_' characters.

Description
Add a short explanation for this role.

Maximum 1000 characters. Use alphanumeric and '+=, @-_' characters.

Step 4: Creating New Service Role For CodeDeploy

- Create a new service role for CodeDeploy and attach **AWSCodeDeployRole** policy which will provide the permissions for our service role to read tags of our EC2 instance, publish information to Amazon SNS topics and much more task.
- Repeat the Above 3 steps again with trusted entity type **AWS Service**, use case **CodeDeploy**.

Select trusted entity Info

Trusted entity type



AWS service

Allow AWS services like EC2, Lambda, or others to perform actions in this account.



AWS account

Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.



SAML 2.0 federation

Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.



Custom trust policy

Create a custom trust policy to enable others to perform actions in this account.

Use case

Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case

CodeDeploy

Choose a use case for the specified service.

Use case



CodeDeploy

Allows CodeDeploy to call AWS services such as Auto Scaling on your behalf.



CodeDeploy for Lambda

Allows CodeDeploy to route traffic to a new version of an AWS Lambda function version on your behalf.



CodeDeploy - ECS

Allows CodeDeploy to read S3 objects, invoke Lambda functions, publish to SNS topics, and update ECS services on your behalf.

- Add **AWSCodeDeployRole** permissions to this creating Role

Add permissions Info

Permissions policies (1) Info

The type of role that you selected requires the following policy.

Policy name

  [AWSCodeDeployRole](#)

► Set permissions boundary - *optional*

- Provide the Name, review and create the role.

Name, review, and create

Role details

Role name

Enter a meaningful name to identify this role.

Maximum 64 characters. Use alphanumeric and '+=, @-_' characters.

Description

Add a short explanation for this role.

Maximum 1000 characters. Use alphanumeric and '+=, @-_' characters.

Step 5: Launch An Linux EC2 instance

- Select the instance with AMI such as "Amazon Linux" and connect to CLI Console.
- Switch to root user from ec2-user to gain admin access power by using following command "sudo su" in Linux.

```
sudo su
```

Step 6: Update The Packages

- The command "sudo yum update" is used in Amazon Linux, CentOS, and Red Hat Linux distributions to update installed packages on your system to their latest available versions.

```
sudo yum update
```

Step 7: Install The Ruby And Wget Software

- The command 'sudo yum install ruby' is used to install the Ruby programming software using the YUM package manager.

```
sudo yum install ruby
```

- The command sudo yum install wget is used to install the "wget" package on a system running Amazon Linux, CentOS, or other Red Hat-based Linux distributions that use the YUM package manager.

```
sudo yum install wget
```

Step 8: Download CodeDeploy Agent Script

- Downloading the AWS CodeDeploy agent installation script from the AWS S3 bucket is an essential step in setting up AWS CodeDeploy for your infrastructure.
- The CodeDeploy agent is a lightweight, scalable software component that enables AWS CodeDeploy to deploy and manage applications on your EC2 instances or on-premises servers.

```
wget
```

```
https://aws-codedeploy-us-east-1.s3.amazonaws.com/latest/install
```

Step 9: Run Installation Script

- The command chmod +x ./install is used to make a file executable in a Unix-like operating system, including Linux.

```
chmod +x ./install
```

The command 'sudo ./install auto' is likely used to run an installation script with superuser (administrator) privileges and pass the "auto" argument to the script.

```
sudo ./install auto
```

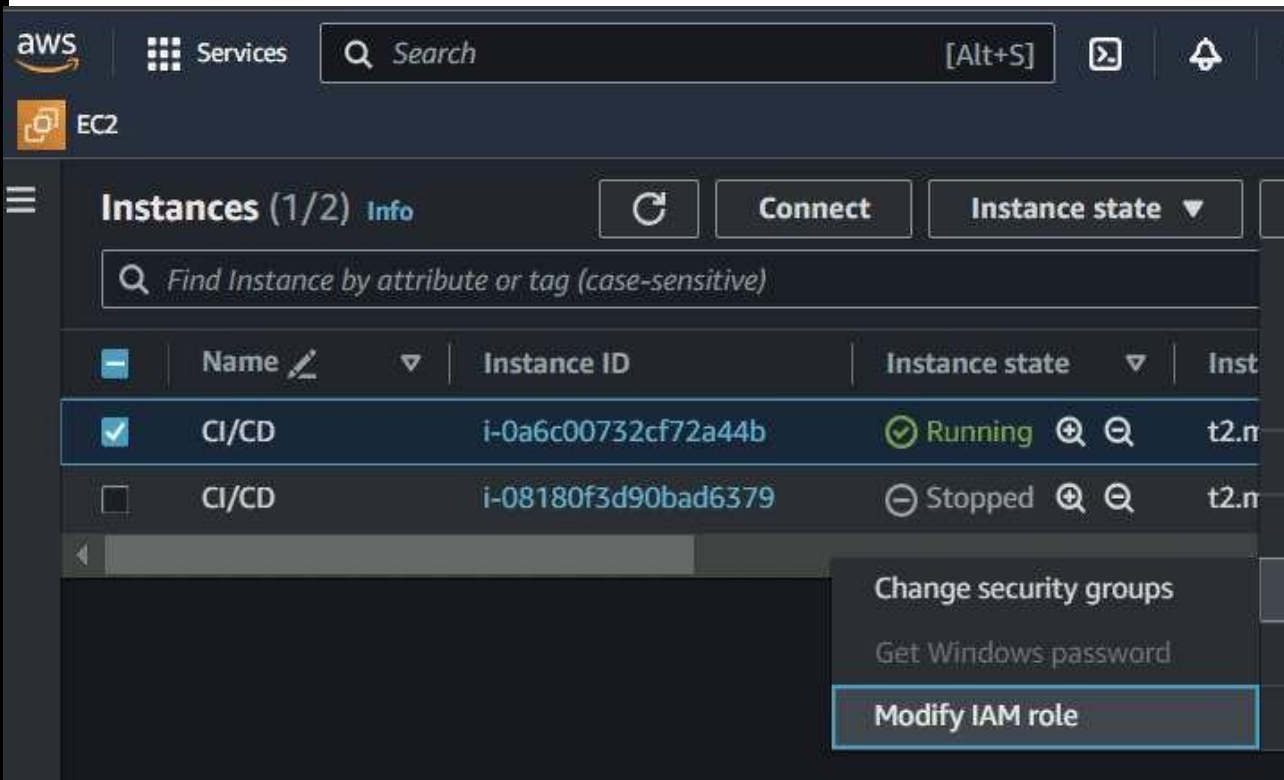
Step 10: Check CodeDeploy Agent Status

- The command sudo service codedeploy-agent status is used to check the status of the AWS CodeDeploy agent running on your system.

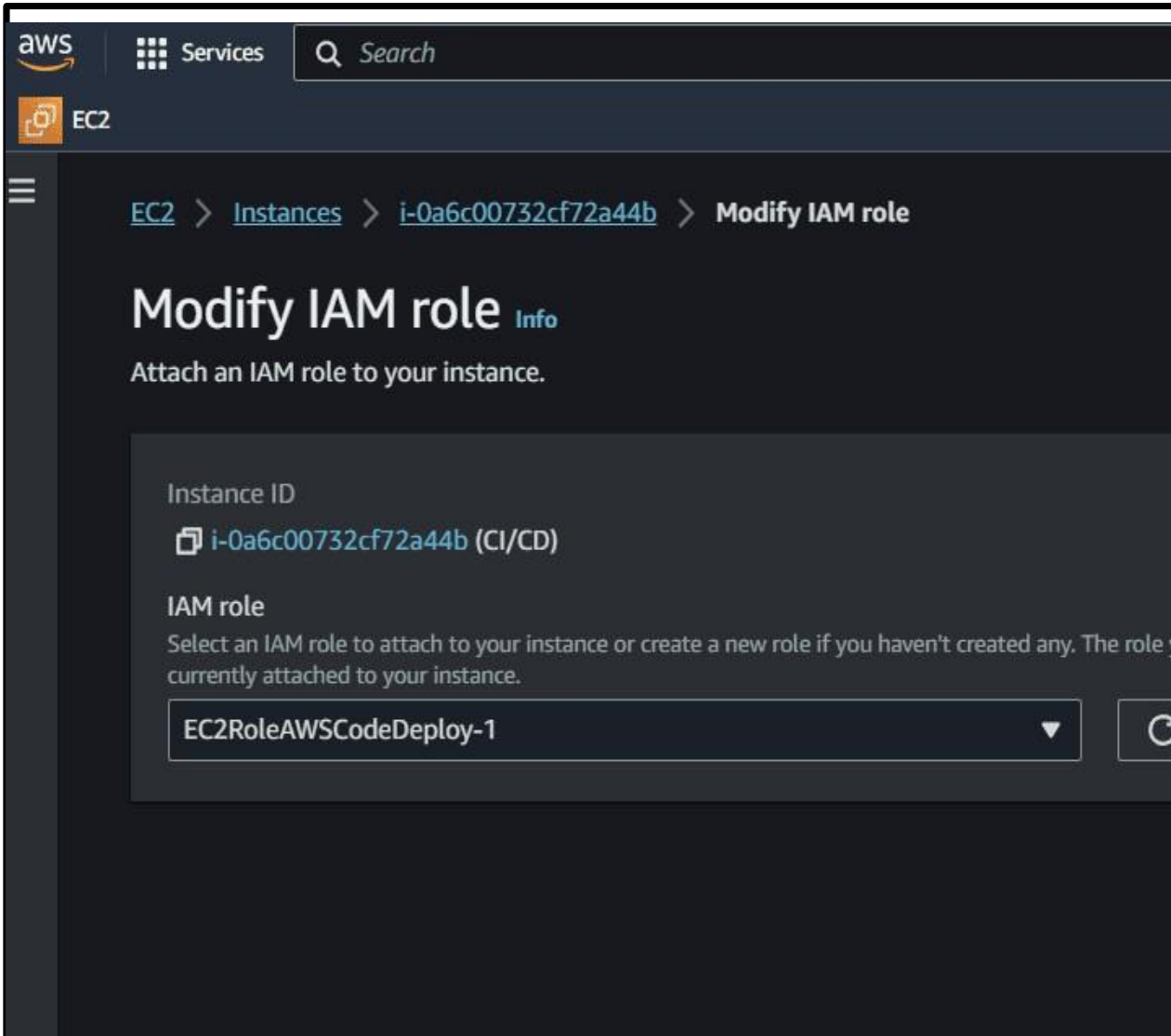
sudo service coddeploy-agent status

Step 11: Modifying IAM Role

- After running the following commands, select the instance and click on "Actions", then click on "Security" and click on "Modify IAM Role". Then choose the above created IAM Role and click on "Update IAM Role".
- After this step, your EC2 instance gets attached with your above created IAM Role.

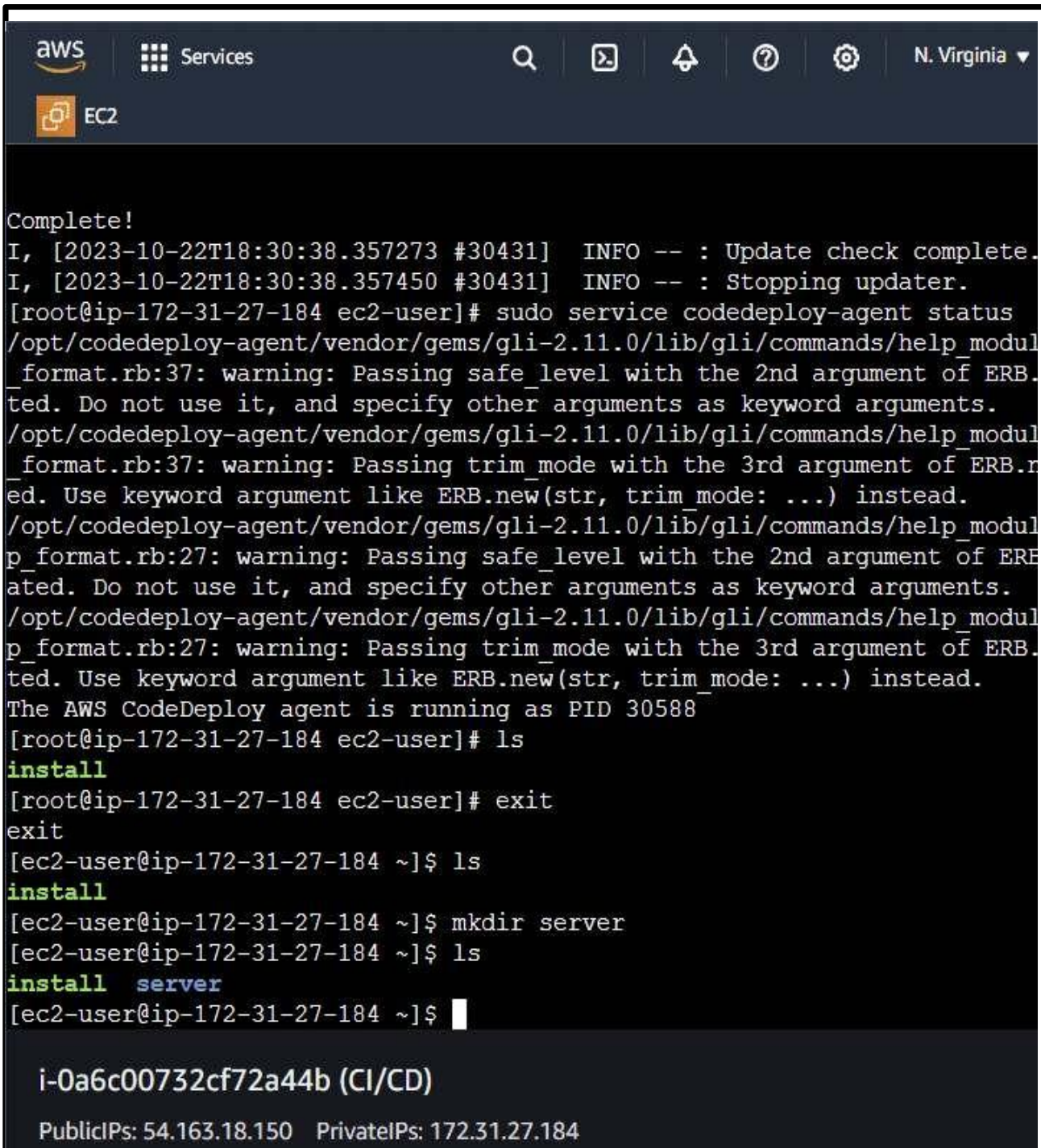


- Modify the IAM role by clicking on the button Update IAM role as shown in the figure.



Step 12: Finalizing The Configuration

After this process, go to the console where your instance is connected and run the command "exit" to exit from the root folder and go back to the EC2 folder. Make a directory on the EC2 folder named "server", this is the directory where my source code will be deployed.



```
aws Services
EC2

Complete!
I, [2023-10-22T18:30:38.357273 #30431] INFO -- : Update check complete.
I, [2023-10-22T18:30:38.357450 #30431] INFO -- : Stopping updater.
[root@ip-172-31-27-184 ec2-user]# sudo service codedeploy-agent status
/opt/codedeploy-agent/vendor/gems/gli-2.11.0/lib/gli/commands/help_modul
_format.rb:37: warning: Passing safe_level with the 2nd argument of ERB.r
ted. Do not use it, and specify other arguments as keyword arguments.
/opt/codedeploy-agent/vendor/gems/gli-2.11.0/lib/gli/commands/help_modul
_format.rb:37: warning: Passing trim_mode with the 3rd argument of ERB.r
ed. Use keyword argument like ERB.new(str, trim_mode: ...) instead.
/opt/codedeploy-agent/vendor/gems/gli-2.11.0/lib/gli/commands/help_modul
p_format.rb:27: warning: Passing safe_level with the 2nd argument of ERB
ated. Do not use it, and specify other arguments as keyword arguments.
/opt/codedeploy-agent/vendor/gems/gli-2.11.0/lib/gli/commands/help_modul
p_format.rb:27: warning: Passing trim_mode with the 3rd argument of ERB.
ted. Use keyword argument like ERB.new(str, trim_mode: ...) instead.
The AWS CodeDeploy agent is running as PID 30588
[root@ip-172-31-27-184 ec2-user]# ls
install
[root@ip-172-31-27-184 ec2-user]# exit
exit
[ec2-user@ip-172-31-27-184 ~]$ ls
install
[ec2-user@ip-172-31-27-184 ~]$ mkdir server
[ec2-user@ip-172-31-27-184 ~]$ ls
install server
[ec2-user@ip-172-31-27-184 ~]$
```

i-0a6c00732cf72a44b (CI/CD)

PublicIPs: 54.163.18.150 PrivateIPs: 172.31.27.184

- Then after doing the above process, come back to the running instances list.
- Select your currently created running instance and go to the "Security" section present at the end of the page.
- Click on the link present under the "Security Groups". After redirecting to the required page, click on "Edit Inbound rules" under the section of "Inbound rules" present at the end of the page.

- Then add a rule, select a port range of your choice and select the source as "Anywhere-IPv4" from the dropdown menu and then click on "Save rules".
- Basically, let me give you a overview what we are actually doing here. In brief, when you add an inbound rule to a security group for an instance with port range (in my case, it was 4000) and set the source to "Anywhere-IPv4," you are allowing any computer or device on the internet to connect to your instance through port 4000.
- This is like opening a door (port 4000) on your server and letting anyone from anywhere access the service or application running on that port.

aws Services Search [Alt+S]

EC2

EC2 > Security Groups > sg-0350c4be16ef18ce1 - launch-wizard-2 > Edit inbound rules

Edit inbound rules [Info](#)

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules [Info](#)

Inbound rule 1

Security group rule ID	Type Info	Protocol Info
sgr-07b0a1e0f832560bc	Custom TCP ▼	TCP
Port range Info	Source type Info	Source Info
4000	Custom ▼	0.0.0.0/
Description - optional Info		

Step 13: Create A New Pipeline

- Create a CodePipeline using Github, CodeBuild and CodeDeploy
- Firstly Create CodePipeline navigate to CodePipeline via AWS Management Console and click on Create pipeline.

Developer Tools > CodePipeline > Pipelines > Create new pipeline

Step 1
Choose pipeline settings

Step 2
Add source stage

Step 3
Add build stage

Step 4
Add deploy stage

Step 5
Review

Choose pipeline settings Info

Pipeline settings

Pipeline name
Enter the pipeline name. You cannot edit the pipeline name after it is created.

reactDemoGithubConnection-1

No more than 100 characters

Service role

☒ **New service role**
Create a service role in your account

Role name

AWSCodePipelineServiceRole-us-east-1-reactDemoGithubConne

Type your service role name

☒ Allow AWS CodePipeline to create a service role so it can be used by the pipeline

► **Advanced settings**

Step 14: Choose Github In Code Source

- After selecting GitHub as the source provider, click on the **Connect to GitHub** button. You'll then be prompt to enter your GitHub login credentials.
- Once you grant AWS CodePipeline access to your GitHub repository, you can select a repository and branch for CodePipeline to upload commits to this repository to your pipeline.

Step 2
Add source stage

Step 3
Add build stage

Step 4
Add deploy stage

Step 5
Review

Source

Source provider
This is where you stored your input artifacts for your pipeline. Choose the provider and then provide the connection details.

GitHub (Version 2)

New GitHub version 2 (app-based) action
To add a GitHub version 2 action in CodePipeline, you create a connection, which uses GitHub App access your repository. Use the options below to choose an existing connection or create a new one [more](#)

Connection
Choose an existing connection that you have already configured, or create a new one and then return to this task.

Q `arn:aws:codestar-connections:us-east-1:160990010622:connection/9757e0f` X or **Connect to new GitHub connection**

Ready to connect
Your GitHub connection is ready for use.

Repository name
Choose a repository in your GitHub account.

Q `DotUrDesign/codepipelinedemo` X

You can type or paste the group path to any project that the provided credentials can access. Use the format 'group/subgroup'.

Branch name
Choose a branch of the repository.

Q `main` X

Change detection options

☒ **Start the pipeline on source code change**
Automatically starts your pipeline when a change occurs in the source code. If turned off, your pipeline only runs if you start it manually or on a schedule.

Output artifact format
Choose the output artifact format.

☒ **CodePipeline default**
AWS CodePipeline uses the default zip format for artifacts in the pipeline. Does not include Git metadata about the repository.

☐ **Full clone**
AWS CodePipeline passes metadata about the repository that allows subsequent actions to do a full Git clone supported for AWS CodeBuild actions.

Cancel Previous

Step 15: Configure CodeBuild (Optional)

- If you haven't created a project prior to creating your pipeline, then you can create a project directly from here by clicking **Create project** button.
- **Note:** Buildspec file is a collection of build commands and related settings, in YAML format, that CodeBuild uses to run a

build. For my project, I created a buildspec.yaml file and added it in the root of my project directory.

Developer Tools > CodePipeline > Pipelines > Create new pipeline

Step 1
Choose pipeline settings

Step 2
Add source stage

Step 3
Add build stage

Step 4
Add deploy stage

Step 5
Review

Add build stage Info

Build - *optional*

Build provider
This is the tool of your build project. Provide build artifact details like operating system, architecture, and platform.

AWS CodeBuild

Region

US East (N. Virginia)

Project name
Choose a build project that you have already created in the AWS CodeBuild console and then return to this task.

reactDemoBuildProject

Environment variables - *optional*
Choose the key, value, and type for your CodeBuild environment variables. In the variable editor, you can add, edit, or delete environment variables. For more information, see [Environment variables in AWS CodePipeline](#). [Learn more](#)

Add environment variable

Build type

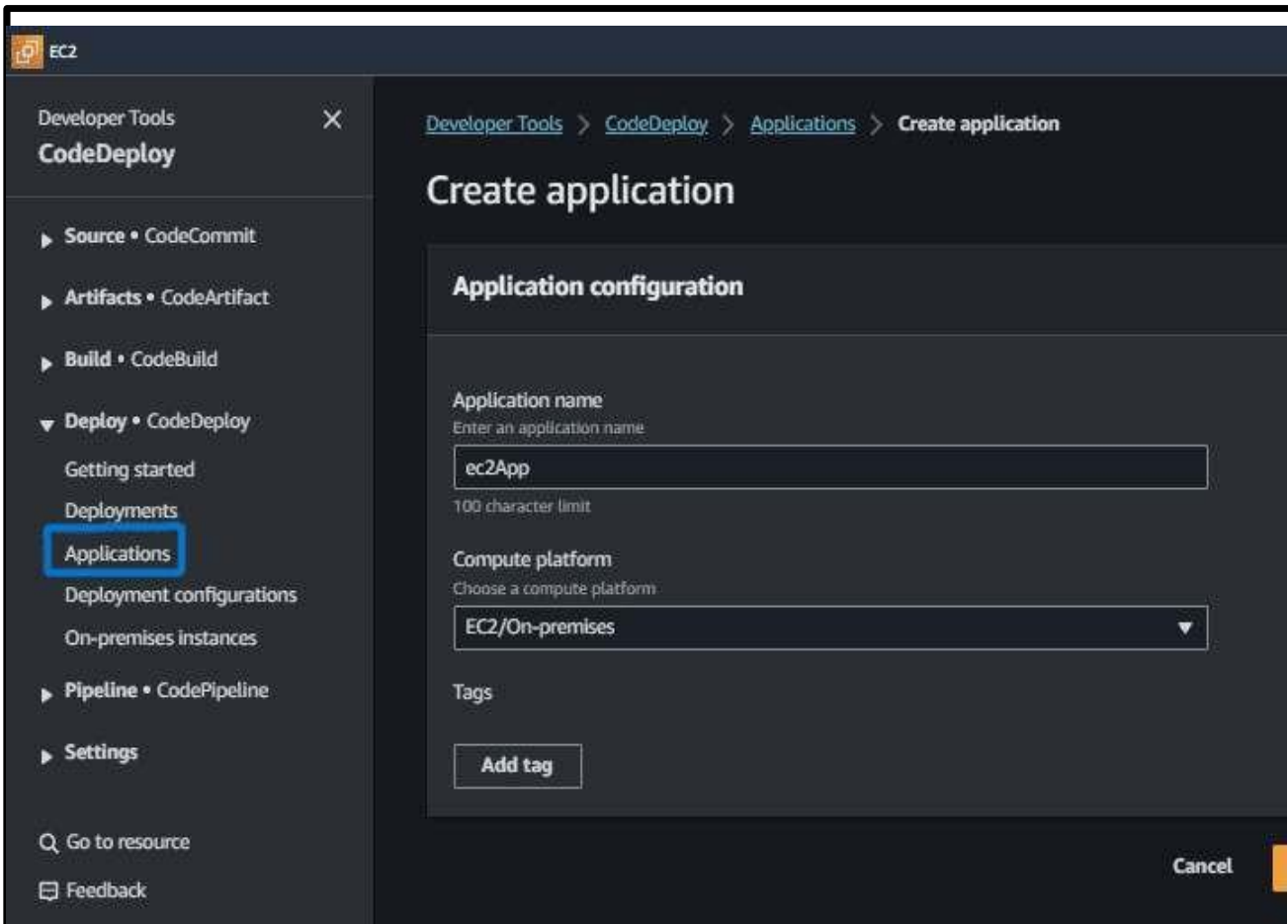
☒ **Single build**
Triggers a single build.

☐ **Batch build**
Triggers multiple build execution.

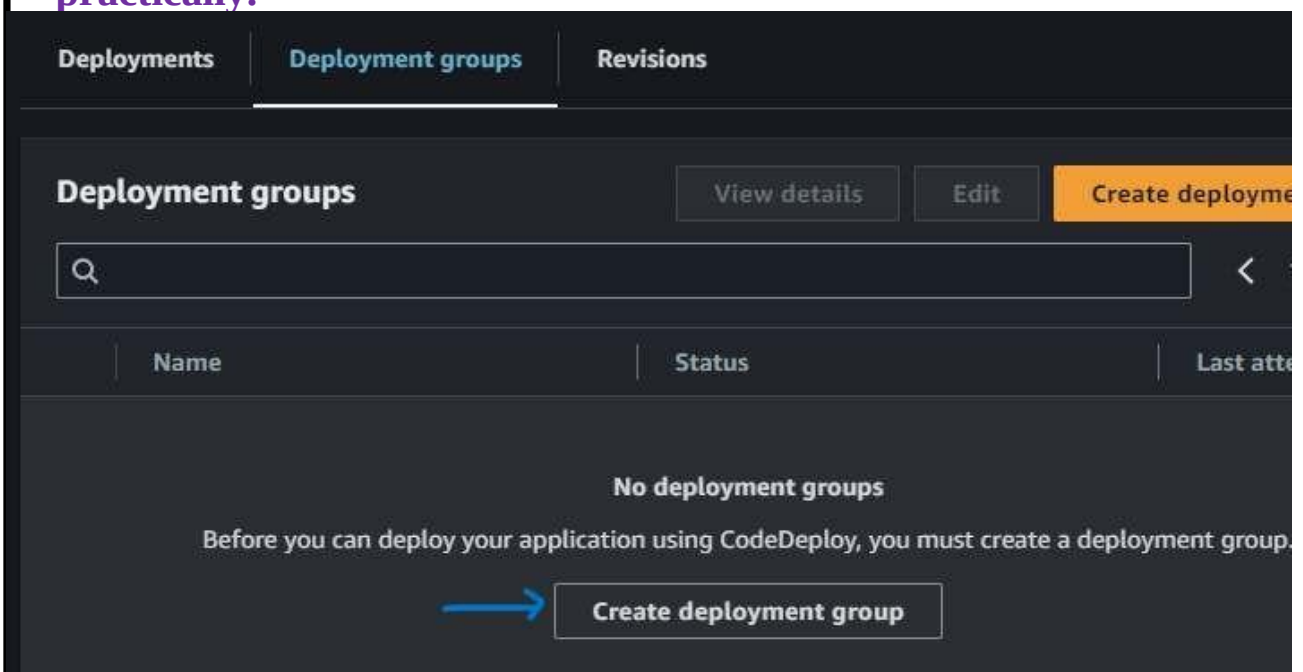
Step 16: Add Deploy Stage

Note : Before going to configure Add Deploy Stage, Let's make duplicate tab of current tab.

- Go to code deploy in the navigation, Select Application, then add create a deployment group.



- Create a deployment Group by clicking on the button "Create deployment group", the following screenshot illustrates with practically.



- **In deployment group Select EC2 instances and select Tag and Value**

Developer Tools > CodeDeploy > Applications > ec2App > Create deployment group

Create deployment group

Application

Application
ec2App
Compute type
EC2/On-premises

Deployment group name

Enter a deployment group name

100 character limit

Service role

Enter a service role
Enter a service role with CodeDeploy permissions that grants AWS CodeDeploy access to your target instances.

Deployment type

Choose how to deploy your application

☒ **In-place**
Updates the instances in the deployment group with the latest application revisions. During a deployment, each instance will be briefly taken offline for its update

☐ **Blue/green**
Replaces the instances in the deployment group with new instances and deploys the latest application revision to them. After instances in the replacement environment are registered with a load balancer, instances from the original environment are deregistered and can be terminated.

- **Provide the Environment configurations such as select the Amazon EC2 Instances and provide the key and values to it.**

Environment configuration

Select any combination of Amazon EC2 Auto Scaling groups, Amazon EC2 Instances, and on-premises instances to add to this deployment

☐ Amazon EC2 Auto Scaling groups

☒ Amazon EC2 instances

2 unique matched instances. [Click here for details](#)

You can add up to three groups of tags for EC2 instances to this deployment group.

One tag group: Any instance identified by the tag group will be deployed to.

Multiple tag groups: Only instances identified by all the tag groups will be deployed to.

Tag group 1

Key

Value - optional

Q Name



Q CI/CD



Remove tag

Add tag

+ Add tag group

☐ On-premises instances

Matching instances

2 unique matched instances. [Click here for details](#)

Agent configuration with AWS Systems Manager [Info](#)



Complete the required prerequisites before AWS Systems Manager can install the CodeDeploy Agent. Make sure the AWS Systems Manager Agent is installed on all instances and attach the required IAM policies to them. [Learn more](#)

Install AWS CodeDeploy Agent

☐ Never

☐ Only once

☒ Now and schedule updates

Uncheck Load Balancer Option

Agent configuration with AWS Systems Manager [Info](#)



Complete the required prerequisites before AWS Systems Manager can install the CodeDeploy Agent. Make sure the AWS Systems Manager Agent is installed on all instances and attach the required IAM policies to them. [Learn more](#)

Install AWS CodeDeploy Agent

☐ Never
☐ Only once
☒ Now and schedule updates

Basic scheduler

Cron expression

14

Days ▼

Deployment settings

Deployment configuration
Choose from a list of default and custom deployment configurations. A deployment configuration is a set of rules that determines how fast an application is deployed and the success or failure conditions for a deployment.

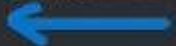
CodeDeployDefault.AllAtOnce ▼

 or

Create deployment configuration

Load balancer

Select a load balancer to manage incoming traffic during the deployment process. The load balancer blocks traffic from each instance while it's being deployed to and allows traffic to it again after the deployment succeeds.

☐ Enable load balancing 

► Advanced - optional

Cancel

Create deployment group

- Finally Come on Add Deploy Stage and select that created Application name & Deployment group

Add deploy stage Info

Deploy - optional

Deploy provider
Choose how you deploy to instances. Choose the provider, and then provide the configuration details for that provider.

AWS CodeDeploy ▼

Region

US East (N. Virginia) ▼

Application name
Choose an application that you have already created in the AWS CodeDeploy console. Or create an application in the AWS CodeDeploy console and then return to this task.

Q ec2App X

Deployment group
Choose a deployment group that you have already created in the AWS CodeDeploy console. Or create a deployment group in the AWS CodeDeploy console and then return to this task.

Q AppDepGrp X

Cancel Previous Skip deploy stage

Step 17: Review And Create

- As a final step review and create it. By creating this we have successfully created a CI/CD pipeline in AWS.

The screenshot displays the AWS CodePipeline console interface. The top navigation bar includes the AWS logo, a search bar, and the user's name 'Pratyush Sahoo' in the N. Virginia region. The left sidebar shows the 'Developer Tools' section with 'CodePipeline' selected. The main content area shows a pipeline execution with three stages: Source, Build, and Deploy, all of which have succeeded.

Source Stage: Succeeded. Pipeline execution ID: eb75cc87-5901-425b-9284-e87a80d9a415. The stage uses GitHub (Version 2) as the provider. It succeeded 22 minutes ago with commit ID 928e84c6. A 'Disable transition' button is visible below the stage.

Build Stage: Succeeded. Pipeline execution ID: eb75cc87-5901-425b-9284-e87a80d9a415. The stage uses AWS CodeBuild as the provider. It succeeded 20 minutes ago. A 'View logs' button is present. A 'Disable transition' button is visible below the stage.

Deploy Stage: Succeeded. Pipeline execution ID: eb75cc87-5901-425b-9284-e87a80d9a415. The stage uses AWS CodeDeploy as the provider.

On the right side of the Build stage, there is a vertical stack of three green checkmarks, indicating the success of the pipeline steps.

PIPELINE CREATED STAGE ALTERNATE METHOD

Developer Tools > CodePipeline > Pipelines > Create new pipeline

Step 1
Choose creation option

Step 2
Choose pipeline settings

Step 3
Add source stage

Step 4
Add build stage

Step 5
Add test stage

Step 6
Add deploy stage

Step 7
Review

Choose creation option [Info](#)

Step 1 of 7

Creation options

Choose one of the following options to create your pipeline.

☐ **Create pipeline from template**
Create a pipeline from a pre-built template for common scenarios.

☒ **Build custom pipeline**
Build a pipeline from scratch to meet your specific needs.

Cancel **Next**

Developer Tools > CodePipeline > Pipelines > Create new pipeline

Step 1
Choose creation option

Step 2
Choose pipeline settings

Step 3
Add source stage

Step 4
Add build stage

Step 5
Add test stage

Step 6
Add deploy stage

Step 7
Review

Choose pipeline settings info

Step 2 of 7

Pipeline settings

Pipeline name

Enter the pipeline name. You cannot edit the pipeline name after it is created.

AWS-DEMO-PIPELINE

No more than 100 characters

Execution mode info

Choose the execution mode for your pipeline. This determines how the pipeline is run.

☐ Superseded

☒ Queued

☐ Parallel

Service role

☒ New service role

Create a service role in your account

☐ Existing service role

Choose an existing service role from your account

Role name

AWSCodePipelineServiceRole-eu-north-1-AWS-DEMO-PIPELINE

Type your service role name

☒ Allow AWS CodePipeline to create a service role so it can be used with this new pipeline

Configure artifact store location, encryption settings, and pipeline variables for your pipeline.

Artifact store

☒ Default location

Create a default S3 bucket in your account.

☐ Custom location

Choose an existing S3 location from your account in the same region and account as your pipeline

Encryption key

☒ Default AWS Managed Key

Use the AWS managed customer master key for CodePipeline in your account to encrypt the data in the artifact store.

☐ Customer Managed Key

To encrypt the data in the artifact store under an AWS KMS customer managed key, specify the key ID, key ARN, or alias ARN.

Variables

You can add variables at the pipeline level. You can choose to assign the value when you start the pipeline. [Learn more](#)

No variables defined at the pipeline level in this pipeline.

Add variable

You can add up to 50 variables.

Cancel

Previous

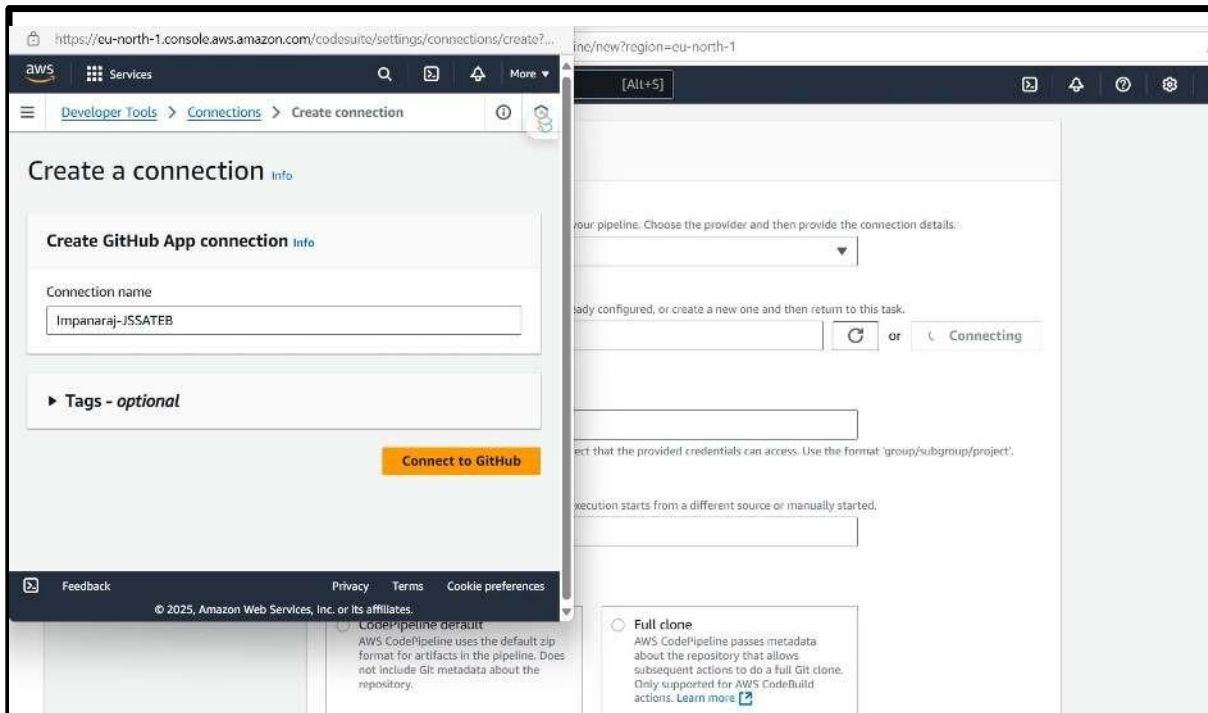
Next

CloudShell Feedback

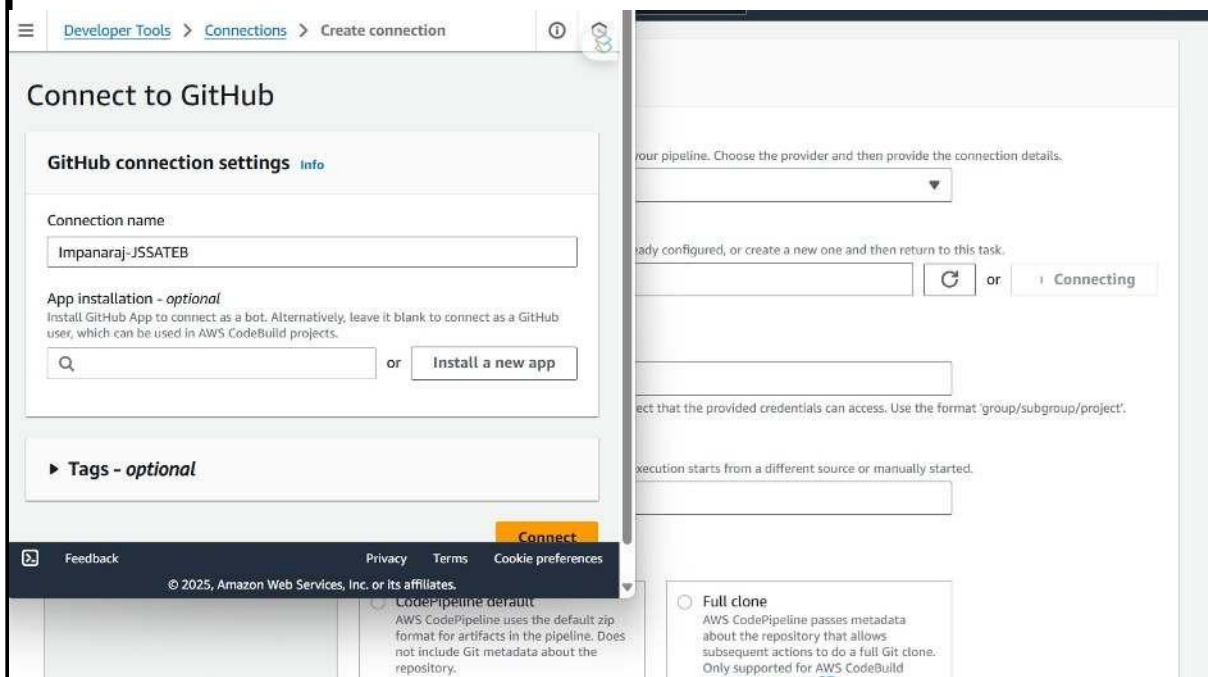
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NEXT

CONNECT TO GITHUB



CONNECT TO GITHUB



CONNECT

Developer Tools > CodePipeline > Pipelines > Create new pipeline

Step 1
Choose creation option

Step 2
Choose pipeline settings

Step 3
Add source stage

Step 4
Add build stage

Step 5
Add test stage

Step 6
Add deploy stage

Step 7
Review

Add source stage Info

Step 3 of 7

Source

Source provider
This is where you store your input artifacts for your pipeline. Choose the provider and then provide the connection details.
GitHub (via GitHub App) ▼

Connection
Choose an existing connection that you have already configured, or create a new one and then return to this task.
arn:aws:codeconnections:eu-north-1:699475937818:connection/42: X ↺ or Connect to GitHub

Repository name
Choose a repository in your GitHub account.
CODEDEPLOY-DEMO

⚠ An unspecified error occurred. Check your network connectivity, and then check to see if there are any issues with the service at the [Service Health Dashboard](#). (Click here to retry)

You can type or paste the group path to any project that the provided credentials can access. Use the format 'group/subgroup/project'.

Default branch

⚠ Not Found (Click here to retry)

Output artifact format
Choose the output artifact format.

☒ CodePipeline default
AWS CodePipeline uses the default zip format for artifacts in the pipeline. Does not include Git metadata about the repository.

☐ Full clone
AWS CodePipeline passes metadata about the repository that allows subsequent actions to do a full Git clone. Only supported for AWS CodeBuild actions. [Learn more](#)

☒ Enable automatic retry on stage failure

Webhook events

Webhook - optional
☒ Start your pipeline on push and pull request events.

▶ Webhook event filters - optional
Starts your pipeline on a specific event.

Remove filters

Cancel

Previous

Next

CloudShell Feedback

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Dept. of CSE, RIT, HASSAN

51

The screenshot shows the AWS CodePipeline console in the 'Add build stage' step (Step 4 of 7). The left sidebar lists the steps: Step 1 (Choose creation option), Step 2 (Choose pipeline settings), Step 3 (Add source stage), Step 4 (Add build stage), and Step 5 (Add deployment stage). The main content area is titled 'Add build stage' and includes the following sections:

- Build - optional**: Build provider section with 'Commands' and 'Other build providers' (selected). The provider is set to 'AWS CodeBuild'.
- Build provider**: Choose the tool you want to use to run build commands and specify artifacts for your build action.
- Build project**: Choose a build project that you have already created in the AWS CodeBuild console. Or create a build project in the AWS CodeBuild console and then return to this task. The search bar contains 'SimpleDockerProject-06d902c306fb' and a 'Create project' button is visible.
- Environment variables - optional**: Choose the key, value, and type for your CodeBuild environment variables. In the value field, you can reference variables generated by CodePipeline. Learn more.
- Add environment variable**: Button to add a new environment variable.
- Build type**: 'Single build' (selected) and 'Batch build' (unselected). 'Single build' triggers a single build, while 'Batch build' triggers multiple builds as a single execution.
- Region**: Set to 'Europe (Stockholm)'.
- Input artifacts**: Choose an input artifact for this action. Learn more. A dropdown menu shows 'SourceArtifact' (Defined by: Source).
- Enable automatic retry on stage failure**: Checked checkbox.

At the bottom, there are buttons for 'Cancel', 'Previous', 'Skip build stage', and 'Next'.

NEW CONSOLE CREATE APPLICATION AND DEPLOYMENT GROUP

The screenshot shows the AWS CodeDeploy console in the 'Applications' page. The left sidebar lists the sections: Source (CodeCommit), Artifacts (CodeArtifact), Build (CodeBuild), Deploy (CodeDeploy), Pipeline (CodePipeline), and Settings. The 'Deploy' section is expanded, showing 'Getting started', 'Deployments', 'Applications' (selected), 'Deployment configurations', and 'On-premises instances'. The main content area is titled 'Applications' and includes the following elements:

- Applications**: Section header with 'Notify', 'View details', 'Deploy application', and 'Create application' buttons.
- Search bar**: A search bar to filter applications.
- Table**: A table listing applications with columns for 'Application name', 'Compute platform', and 'Created'.
- Table Data**: One application is listed: 'ABC-DEMO' with 'EC2/On-premises' as the compute platform and '1 minute ago' as the creation time.

At the bottom, there are buttons for 'Go to resource' and 'Feedback'.

The screenshot displays the AWS CodeDeploy console interface. The top navigation bar includes the AWS logo, 'Services', a search bar, and regional information (Europe (Stockholm)). The left sidebar shows the 'Developer Tools' section with 'CodeDeploy' selected. The main content area is titled 'ABC-DEMO' and features a 'Notify' button and a 'Delete application' button. Below this, the 'Application details' section shows the application name 'ABC-DEMO' and the compute platform 'EC2/On-premises'. The 'Deployment groups' tab is active, displaying a message: 'No deployment groups. Before you can deploy your application using CodeDeploy, you must create a deployment group.' A 'Create deployment group' button is visible at the bottom of this section. The footer of the console shows 'CloudShell', 'Feedback', and copyright information for Amazon Web Services, Inc.

Developer Tools > CodePipeline > Pipelines > Create new pipeline

Step 1
Choose creation option

Step 2
Choose pipeline settings

Step 3
Add source stage

Step 4
Add build stage

Step 5
Add test stage

Step 6
Add deploy stage

Step 7
Review

Review info

Step 7 of 7

Step 2: Choose pipeline settings

Pipeline settings

Pipeline name

AWS-DEMO-PIPELINE

Pipeline type

V2

Execution mode

QUEUED

Artifact location

codepipeline-eu-north-1-249305857888

Service role name

AWSCodePipelineServiceRole-eu-north-1-AWS-DEMO-PIPELINE

Step 3: Add source stage

Source action provider

Source action provider

AWS CodeStarSourceConnection

OutputArtifactFormat

CODE_ZIP

DetectChanges

true

ConnectionArn

arn:aws:codeconnections:eu-north-1:699475937818:connection/4223c4da-d25c-4599-853e-3e6130470f26

FullRepositoryId

CODEDEPLOY-DEMO

Step 6: Add deploy stage

Deploy action provider

Deploy action provider

AWS CodeDeploy

ApplicationName

ABC-DEMO

DeploymentGroupName

ABC-DG

Configure automatic rollback on stage failure

Enabled

Enable automatic retry on stage failure

Disabled

Cancel

Previous

Create pipeline

aws

Services

CODEDEploy

×

📄 🔔 ⌛ ⚙️

Europe (Stockholm) ▼

Idpounce ▼

MAIN

Enable automatic retry on stage failure
Enabled

Step 6: Add deploy stage

Deploy action provider

Deploy action provider
AWS CodeDeploy

ApplicationName
ABC-DEMO

DeploymentGroupName
ABC-DG

Configure automatic rollback on stage failure
Enabled

Enable automatic retry on stage failure
Disabled

Cancel Previous Create pipeline

CloudShell Feedback

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